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Single-phase incomplete multi-view clustering via joint anchor refactoring and tensorial bipartite graph learning

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ABSTRACT

The inherent incompleteness of multi-view data poses significant challenges for conventional clustering methods, particularly those based on bipartite graph construction. Although existing approaches have made notable progress by refining graph architecture, they often overlook the critical role of anchor quality in determining overall clustering performance. To address this limitation, we propose a Single-phase Incomplete Multi-view Clustering via Joint Anchor Refactoring and Tensorial Bipartite Graph (SIMC-ARTB), a novel unified framework that jointly integrates anchor refactoring, aligned graph fusion, and single-phase label learning. Specifically, SIMC-ARTB first learns high-quality anchors through sample projection matrices and introduces a novel anchor refactoring strategy to enforce distributional consistency between anchors and original data. To effectively fuse incomplete view-specific bipartite graphs, we further propose an alignment-based fusion method that incorporates orthogonal transformation matrices and adaptive weights, yielding a consistent and well-aligned graph representation. To capture high-order structural correlations across multiple views, we formulate a weighted tensor nuclear norm regularization term, which enhances the robustness and low-rank consistency of the fused graph. Moreover, departing from the widely adopted two-stage paradigm, SIMC-ARTB embeds discrete label learning directly into the optimization process, enabling mutual reinforcement between consistency graph learning and discrete label assignment. Extensive experimental results on benchmark datasets demonstrate that SIMC-ARTB consistently outperforms state-of-the-art incomplete multi-view clustering methods in terms of clustering accuracy, stability, and scalability.

1. Introduction

Multi-view Clustering (MC) has gained attention for leveraging complementary information from multiple feature representations (Fang et al., 2023; Qin et al., 2025; Yan et al., 2021). Integrating heterogeneous views improves unsupervised learning in areas like medical diagnosis (Krones et al., 2025), social media analytics (Yang et al., 2023), and identity recognition (Xing & Zhao, 2024). However, real-world multi-view data are often incomplete due to sensor failures, data corruption, privacy constraints, or high costs (Tang et al., 2024; Wen et al., 2022), posing challenges for conventional MC methods, especially those using bipartite graph-based modeling.

To address scalability and efficiency issues in graph-based clustering, anchor-based bipartite graph models have been widely adopted in Incomplete Multi-view Clustering (IMVC) (Liu et al., 2022; Wen et al., 2023). By introducing a relatively small set of anchors, these meth-

ods significantly reduce computational complexity while preserving essential structural information for clustering. Existing IMVC approaches have primarily focused on refining bipartite graph architectures, such as adaptive graph learning, view-weighted fusion, and graph completion strategies (Li et al., 2025; Zhao et al., 2023). Despite their effectiveness, most of these methods implicitly assume that anchors are reliable and representative. In practice, however, anchor quality is highly sensitive to view incompleteness, leading to biased or noisy anchor representations that propagate errors through the bipartite graph and fundamentally limit clustering performance. Anchors constructed directly from incomplete features are prone to deviating from the true data prototypes, thereby weakening the discriminative power of anchor-sample relationships. Meanwhile, missing instances may include highly representative samples that contribute significantly to describing cluster structures. The absence of such critical samples will inevitably bias the estimation of the underlying data distribution, thereby impairing the accuracy of

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anchor learning. This critical issue of anchor quality degradation has received limited attention in current IMVC research (He et al., 2025).

In addition to anchor reliability, incomplete multi-view clustering faces further challenges. Due to heterogeneous feature spaces and distinct missing patterns across views, view-specific bipartite graphs often exhibit significant structural misalignment, making direct fusion ineffective or even detrimental (Yao et al., 2025). Moreover, the majority of existing methods adopt a two-stage learning paradigm, in which graph construction and clustering label assignment are performed separately. Such decoupled optimization restricts interaction between representation learning and clustering decisions, preventing mutual reinforcement and limiting overall performance (Wu et al., 2026). Furthermore, effectively capturing high-order consistency across multiple incomplete views remains difficult under noisy and partially observed conditions (Long et al., 2025; Wang et al., 2025; Yin et al., 2025).

Motivated by these challenges, this paper addresses the fundamental problem of how to perform effective incomplete multi-view clustering when anchor representations are unreliable, view-specific graph structures are misaligned, and graph learning is decoupled from label inference. To this end, we propose Single-phase Incomplete Multi-view Clustering via Joint Anchor Refactoring and Tensorial Bipartite Graph (SIMC-ARTB), a unified framework that jointly enhances anchor quality, aligns and fuses incomplete bipartite graphs, and learns clustering labels within a single optimization process.

Specifically, SIMC-ARTB introduces a novel anchor refactoring strategy that learns high-quality anchors via sample projection matrices, explicitly enforcing distributional consistency between anchors and the original data to mitigate the effects of view incompleteness. To integrate incomplete view-specific bipartite graphs, we develop an alignment-based fusion mechanism incorporating orthogonal transformation matrices and adaptive view weights, enabling structural alignment and balanced information aggregation across heterogeneous views. To capture high-order cross-view correlations and improve robustness, we apply a weighted tensor nuclear norm regularization to the fused bipartite graphs, promoting global low-rank consistency while suppressing noise induced by missing data. Finally, moving beyond the conventional two-stage paradigm, SIMC-ARTB embeds discrete label learning directly into the optimization objective, allowing consistency graph learning and clustering label assignment to be jointly refined within a single-phase framework. An efficient optimization strategy is also proposed to solve the model with relatively low computational complexity.

The primary contributions of this paper are summarized as follows:

- We propose a novel anchor refactoring strategy that enforces distributional consistency between anchors and original data, significantly improving anchor reliability and diversity under incomplete views.
- We develop an aligned tensorial bipartite graph fusion framework that integrates orthogonal transformations, adaptive view weighting, and weighted tensor nuclear norm regularization to capture high-order cross-view consistency.
- We design a single-phase joint optimization paradigm that tightly couples consistency graph learning with discrete label assignment, overcoming the limitations of conventional two-stage IMVC methods.
- We conduct extensive experiments on benchmark incomplete multi-view datasets, demonstrating that SIMC-ARTB consistently outperforms state-of-the-art methods in terms of clustering accuracy, stability, scalability, and robustness.

The remainder of the paper is structured as follows: Section 2 reviews related work and preliminaries. Sections 3 and 4 present the proposed approach and its corresponding optimization framework, respectively. Comprehensive experimental evaluations are reported in Section 5. Finally, Section 6 concludes the paper and outlines future research directions.

Table 1

Definition of the basic notations.

Symbols	Meaning
$I, \mathbf{1}, \mathbf{0}$	Identity, all-ones, all-zeros matrix
V, n, m, c	Counts of views, samples, anchors, clusters
n_v	Counts of complete samples of v -th view
d_v	Dimensionality of v -th view
$\omega \in \mathbb{R}^{V \times 1}$	View weights vector
$X^{(v)} \in \mathbb{R}^{d_v \times n}$	Observed samples of v -th view
$M^{(v)} \in \mathbb{R}^{n \times n_v}$	Mask matrix corresponding to v -th view
$U^{(v)} \in \mathbb{R}^{d_v \times m}$	Anchor matrix of the v -th view
$B^{(v)} \in \mathbb{R}^{m \times n}$	Bipartite graph of the v -th view
$F^{(v)} \in \mathbb{R}^{n_v \times m}$	Projection matrix of the v -th view
$G^{(v)} \in \mathbb{R}^{m \times m}$	Alignment matrix of v -th view
$P \in \mathbb{R}^{m \times n}$	Consistent bipartite graph
$H \in \mathbb{R}^{m \times c}$	Clustering centers matrix
$Y \in \mathbb{R}^{c \times n}$	Clustering labels graph
$S \in \mathbb{R}^{m \times n \times (V+1)}$	Third-order tensor

2. Related work and preliminaries

This section presents a review of recent advances in incomplete multi-view clustering, including bipartite graph-guided IMC, tensor-based IMC, and tensor operator preliminaries. The symbols and notations used throughout this paper are summarized in Table 1.

2.1. Bipartite graph guided IMC

Bipartite Graph (BG) learning has been extensively studied as an efficient alternative to full graph modeling (Nie et al., 2023; Zhang et al., 2023). Its central objective is to select a subset of representative anchors and construct a similarity graph that encodes the relationships between anchors and original data points (Li et al., 2024). By avoiding exhaustive pairwise similarity calculations, the construction of a bipartite graph substantially reduces the learning complexity, making it particularly suitable for large-scale IMC tasks (Zhao et al., 2023). Formally, let $X^{(v)} \in \mathbb{R}^{d_v \times n}$ denote an incomplete multi-view dataset with n samples and c clusters, and let $M^{(v)} \in \mathbb{R}^{n \times n_v}$ represent the associated missing mask matrix. The general learning framework is then defined as follows:

$$\min_{B^{(v)}, U^{(v)}, P} \sum_{v=1}^V \mathcal{L}_1(U^{(v)}, B^{(v)}) + \mathcal{L}_2(B^{(v)}, P) \quad (1)$$

s.t. $B^{(v)} \geq 0, B^{(v)} \mathbf{1} = \mathbf{1}, U^{(v)T} U^{(v)} = I_m.$

For model (1), $\mathcal{L}_1(U^{(v)}, B^{(v)})$ corresponds to the bipartite graph learning module, $\mathcal{L}_2(B^{(v)}, P)$ presents fusion module, while $\mathcal{L}_1(\cdot)$ and $\mathcal{L}_2(\cdot)$ denote the loss functions for bipartite graph learning and bipartite graph fusion, respectively. The common choices for these loss functions include the squared Frobenius norm $\|\cdot\|_F^2$ and the $l_{2,1}$ -norm $\|\cdot\|_{2,1}$. Here, $B^{(v)} \in \mathbb{R}^{m \times n}$ represents the bipartite graph of the v -th view, and $P \in \mathbb{R}^{m \times n}$ denotes the complete consistent bipartite graph. $U^{(v)} \in \mathbb{R}^{d_v \times m}$ denotes the anchor matrix for the v -th view, which consists of m anchor points, with each column corresponding to one anchor. To obtain the anchors, Singular Value Decomposition (SVD) is employed under Orthogonality Constraint (OC), whereas matrix factorization is adopted in the unconstrained settings. Motivated by anchor diversity, Yu et al. (2024b) employed different numbers of anchors for each view to construct multi-scale bipartite graphs, enabling efficient IMC. Subsequently, Yu et al. (2024a) introduced anchors to facilitate communication across latent spaces and constructed a view-shared anchor graph. Later, Shen et al. (2025) incorporated entropy-induced reliable anchor learning with graph construction to reveal the underlying cluster information. However, these prior works emphasize anchor diversity in isolation, which inherently constrains the representational capacity of anchors with respect to the intrinsic data distribution.

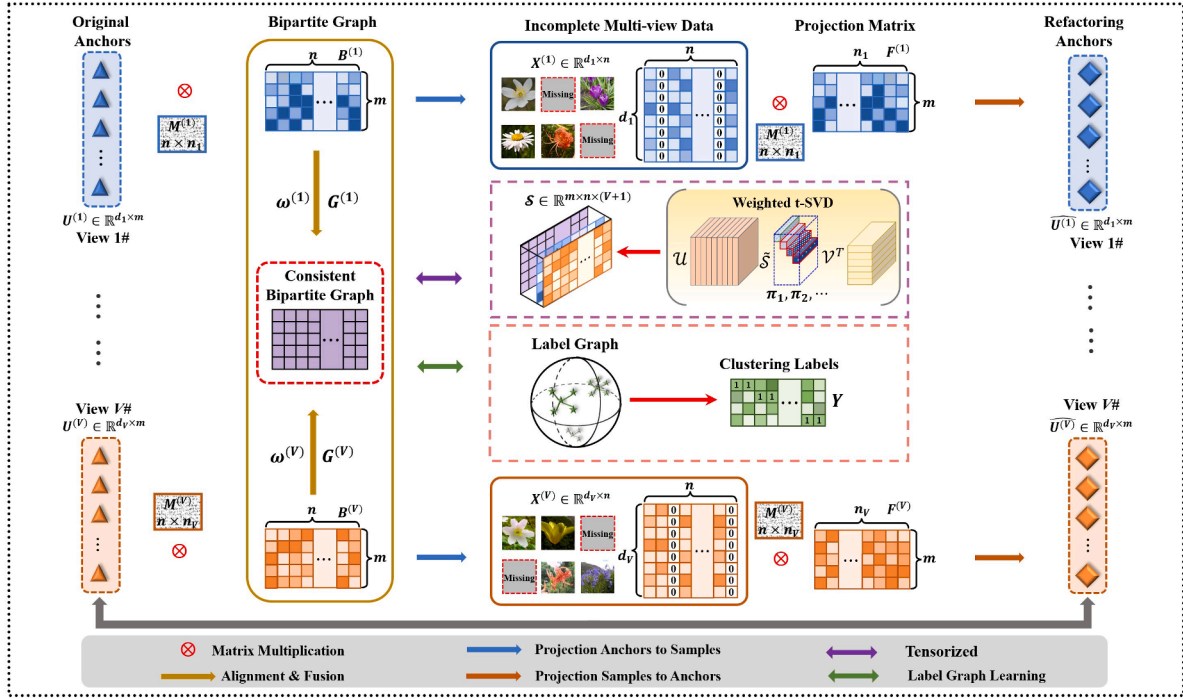


Fig. 1. Overview of the SIMC-ARTB architecture: For each incomplete view, SIMC-ARTB first constructs a bipartite graph $B^{(v)}$ using the incomplete samples together with the original anchors. A sample projection matrix $F^{(v)}$ is then introduced to refine the anchors, thereby enhancing anchor representations. Subsequently, a consistent bipartite graph learning strategy is employed to fuse incomplete graphs via adaptive weighting and alignment matrices. Furthermore, a weighted tensor nuclear norm is imposed on the S to capture high-order inter-view correlations and facilitate missing information recovery. Finally, label graph learning is designed to directly derive clustering label assignment. Overall, SIMC-ARTB integrates anchor refactoring, consistent bipartite graph learning, and label graph learning within a unified framework, enabling these components to mutually reinforce each other and ultimately improve clustering performance.

2.2. Tensor-based IMC

Tensor-based IMC methods have received considerable interest for their capacity to model and recover missing inter-view relationships, enabling the discovery of complex structural interactions and latent representations across different views (Li et al., 2025; Xia et al., 2022). Let $Z^{(v)} \in \mathbb{R}^{n \times n}$ or $Z^{(v)} \in \mathbb{R}^{m \times n}$ denote the incomplete correlation graph. In general, the formulation of tensor-based with graph learning methods is as follows:

$$\min_{Z^{(v)}, Z} \sum_{v=1}^V \mathcal{J}_1(Z^{(v)}, Z) + \mathcal{J}_2(Z) \quad (2)$$

$$s.t. \quad Z = \Phi([Z^{(1)}; Z^{(2)}; \dots; Z^{(V)}]),$$

where $\mathcal{J}_1(\cdot)$ denotes a function for the consistent correlation graph learning. The $\Phi(\cdot)$ indicates an operator for building a 3-order tensor. Then, $\mathcal{J}_2$ is the tensor norm constraints, including Tensor Nuclear Norm (TNN) $\|\cdot\|_{\Theta}$, Weighted Tensor Nuclear Norm (WTNN) $\|\cdot\|_{\pi, \Theta}$, tensor Schatten p -norm ($\mathcal{S}p$) $\|\cdot\|_{\mathcal{S}p}^p$, and other low-rank tensor norm with non-convex rank function (Ji & Feng, 2025). Guo et al. (2025) propose a basic indicator matrix-based late fusion framework for IMC, which constructs the 3-order base tensor to explore the higher-order correlations. Chen et al. (2023) exploits a integrates dynamic anchor learning and tensor modeling for scalable IMC, which jointly capture intra-view structures and high-order inter-view correlations. Cai et al. (2026) devise an aligned anchor graph learning method for IMC, which imposes tensor-Singular Value Decomposition (t-SVD) based tensor nuclear norm regularization on the anchor graph tensor with a binary alignment matrix. However, this method only considers the alignment of the incomplete bipartite graph to the dominant view while disregarding the means of a consistent bipartite graph for high-order correlation learning.

2.3. Tensor operation preliminaries

Definition 1 (t-SVD Gao et al., 2020). Given a third-order tensor $\mathcal{W} \in \mathbb{R}^{\eta_1 \times \eta_2 \times \eta_3}$, the tensor-Singular Value Decomposition (t-SVD) operation can be defined as

$$\mathcal{W} = \mathcal{U} * \mathcal{O} * \mathcal{V}^T, \quad (3)$$

where $\mathcal{U} \in \mathbb{R}^{\eta_1 \times \eta_1 \times \eta_3}$ and $\mathcal{V} \in \mathbb{R}^{\eta_2 \times \eta_2 \times \eta_3}$ are subject to tensor orthogonality constraints, i.e., $\mathcal{U}^T * \mathcal{U} = \mathcal{I}$ and $\mathcal{V}^T * \mathcal{V} = \mathcal{I}$. Here, $\mathcal{O}$ denotes an f -diagonal tensor, and $(*)$ denotes the tensor product.

Definition 2 (WTNN Gao et al., 2020). Let a tensor $\mathcal{W} \in \mathbb{R}^{\eta_1 \times \eta_2 \times \eta_3}$, the Weighted Tensor Nuclear Norm (WTNN) operation is formulated as

$$\|\mathcal{W}\|_{\pi, \Theta} = \sum_{t=1}^{\eta_3} \|\widehat{\mathcal{W}}^{(t)}\|_{\pi, \Theta} = \sum_{t=1}^{\eta_3} \sum_{r=1}^{\min(\eta_1, \eta_2)} \pi_r \sigma_r(\widehat{\mathcal{W}}^{(t)}), \quad (4)$$

where $\widehat{\mathcal{W}}^{(t)}$ denotes the t -th frontal slice of $\mathcal{W}$ in the Fourier domain, $\sigma_r(\widehat{\mathcal{W}}^{(t)})$ denotes its r -th largest singular value, and π_r is the corresponding weight.

3. The proposed model

In this section, we present a unified framework that integrates refined anchor refactoring, consistent bipartite graph learning, and label graph learning derived from the consistent bipartite graph. To facilitate understanding of the proposed approach, the overall workflow is illustrated in Fig. 1.

3.1. Refined anchor refactoring

In conventional incomplete multi-view clustering frameworks based on bipartite graphs, the primary focus has typically been on constructing the bipartite graph structure, while the influence of anchor quality

on the overall representation capacity is often overlooked. However, the quality of anchors plays a crucial role in determining the effectiveness of the learned graph representation. To address this limitation, we introduce a coupled anchor refactoring-based bipartite graph learning approach, which aims to enhance the discriminative capability of the graph by improving the representational quality of anchors. Specifically, the proposed anchor refinement strategy projects the original anchors into the incomplete multi-view feature space via a bipartite graph, thereby generating a more representative set of anchors. This process can be explicitly expressed as follows:

$$\begin{aligned} \min_{B^{(v)}, U^{(v)}} \sum_{v=1}^V \left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2 \\ \text{s.t. } M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}, \end{aligned} \quad (5)$$

where $M^{(v)} = I_{n \times n}(:, \eta^{(v)}) \in \{0, 1\}^{n \times n_v}$ is a mask matrix. And $\eta^{(v)}$ is defined as the ordered index vector of samples observed in the v -th view. $B^{(v)}$ denotes the bipartite graph, and $U^{(v)}$ represents the original anchor matrix. By discouraging highly correlated sample representations and encouraging diverse sample-anchor relationships, $M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}$ constraint allows the consensus anchor graph to better capture the intrinsic heterogeneity of the multi-view data.

To ensure that these anchors accurately reflect the underlying distribution of the original data, an anchor refactoring mechanism is incorporated, encouraging the learned anchors to closely align with the intrinsic features of the input samples. Finally, a projection matrix is employed to map the multi-view features back to the original anchor space, resulting in more refined and expressive anchor representations. The formulation is given as follows:

$$\widehat{U^{(v)}} \leftarrow X^{(v)} M^{(v)} F^{(v)}, \quad (6)$$

where $F^{(v)}$ refers to the projection matrix under $F^{(v)T} F^{(v)} = I_m$, and $\widehat{U^{(v)}}$ denotes its refactored anchors matrix. Considering the information fidelity of anchors before and after refactoring. It can be shown that

$$\widehat{U^{(v)}} = U^{(v)} \Rightarrow X^{(v)} M^{(v)} F^{(v)} = U^{(v)} \quad (7)$$

Since the original anchors $U^{(v)}$ are learned from incomplete samples via $\left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2$, their representations inevitably contain estimation bias introduced by missing views. The refactored anchors are obtained as $X^{(v)} M^{(v)} F^{(v)} = \widehat{U^{(v)}}$, where the anchor representations are able to better capture the underlying cluster structure and provide more reliable anchor-sample relationships. Furthermore, the consistency regularization with $\widehat{U^{(v)}} = U^{(v)}$ preserves useful information from the original anchors and prevents the refinement process from introducing excessive deviations. This joint design effectively improves anchor quality and enhances the robustness of anchor learning in incomplete multi-view scenarios.

3.2. Consistent bipartite graph learning

Constructing a unified and complete bipartite graph from incomplete multi-view data requires a learning framework capable of addressing both structural alignment and robust integration. This process begins with the alignment of individual view-specific bipartite graphs, which may exhibit divergent cluster distributions due to missing data. To harmonize these distributions, an alignment matrix $G^{(v)}$ is introduced to enforce a coherent geometric structure across views prior to fusion. The integration is then performed via an adaptive weighting scheme, which dynamically adjusts the contribution of each view to the consistent bipartite graph, thereby mitigating the influence of unreliable or incomplete views. To avoid weight collapse and ensure balanced contributions across views, the sum of the squared weights is constrained to be equal

to 1. The resulting formulation is given as follows:

$$\begin{aligned} \max_{P, G^{(v)}, \omega^{(v)}} \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] \\ \text{s.t. } M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}, G^{(v)T} G^{(v)} = I_m, \\ P P^T = I_m, \omega^{(v)} \geq 0, \|\omega\|_2^2 = 1, \end{aligned} \quad (8)$$

where $G^{(v)}$ denotes the alignment matrix, $\omega^{(v)}$ represents adaptive weight associated with the v -th view, and P denotes the learned consistent bipartite graph.

To further enhance the quality of the consistent structure, the entire set of view-specific graphs $B^{(v)}$ and the fused graph P are organized into a unified tensor representation. Specifically, we stack all $B^{(v)}$ and P into a third-order tensor $S \in \mathbb{R}^{m \times n \times (V+1)}$ to promote cross-view complete consistency based on the learned representations.

$$\begin{aligned} \max_{P, G^{(v)}, \omega^{(v)}} \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] - \gamma \|S\|_{\pi, \Theta} \\ \text{s.t. } M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}, G^{(v)T} G^{(v)} = I_m, \\ P P^T = I_m, \omega^{(v)} \geq 0, \|\omega\|_2^2 = 1, \\ S = \Psi \left(\left[B^{(1)}; \dots; B^{(V)}; P \right] \right). \end{aligned} \quad (9)$$

Here, $\Psi(\cdot)$ constructs a unified tensor by stacking consistent and multi-view bipartite graph. $\|S\|_{\pi, \Theta}$ denotes the weighted tensor regularization term, which is applied as a high-order consistency constraint, adaptively leveraging the latent correlations within this multi-way array.

The above approach Eq. (8) recovers a complete bipartite graph by directly fusing the bipartite graphs from multiple views. Such a strategy primarily relies on the explicit integration of graph structures across views and can therefore be regarded as a low-order, explicit recovery mechanism. In contrast, tensor-based consistent bipartite graph learning is capable of capturing the latent and implicit data distribution patterns shared among multiple views, thereby forming a high-order, implicit learning paradigm. Furthermore, embedding the consistent bipartite graph within a tensor framework not only enables the effective propagation of global consistency information to incomplete bipartite graphs but also facilitates iterative optimization by leveraging the latent structural relationships contained within these incomplete graphs.

3.3. Label graph learning from a consistent bipartite graph

To circumvent the instability and suboptimality inherent in the widely adopted two-stage approach for IMC, we introduce a principled approach that performs one-step label learning directly from the complete and consistent bipartite graph. Conventional methods typically apply *Kmeans* clustering to the learned consistent representation. However, this strategy suffers from two key limitations: first, the stochastic nature of *Kmeans* introduces instability in the clustering results; second, the separation between representation learning and label assignment often leads to suboptimal clustering performance. To address these issues, we reformulate the final clustering objective. Instead of treating the learned graph as an intermediate representation followed by a separate clustering stage, we directly construct a label graph from the unified bipartite graph, enabling label assignment to be learned jointly with the representation. The formulation is given as follows:

$$\begin{aligned} \min_Y \sum_{i=1}^c \sum_{j=1}^n y_{ij} \left\| H_{(:,i)} - P_{(:,j)} \right\|_2^2 \\ \text{s.t. } H^T H = I_c, P P^T = I_m, y_{ij} \in \{0, 1\}, \end{aligned} \quad (10)$$

where Y denotes the clustering label matrix, and H represents the corresponding cluster centers. The independence constraint is imposed on the learned cluster centers via $H^T H = I_c$. This regularization prevents

different cluster prototypes from converging to similar directions in the feature space and promotes a more balanced distribution of cluster prototypes. Consequently, the learned clusters become more distinguishable, and samples are less likely to be concentrated in only a few clusters.

By integrating all stages, the final objective function of SIMC-ARTB can be explicitly formulated as follows:

$$\begin{aligned}
\min_{\Phi} & \underbrace{\sum_{v=1}^V \left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2}_{\text{Incomplete bipartite graph learning}} - \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] \\
& \underbrace{+ \beta \sum_{i=1}^c \sum_{j=1}^n y_{ij} \left\| H_{(:,i)} - P_{(:,j)} \right\|_2^2}_{\text{Label graph learning}} + \gamma \|S\|_{\pi, \Theta} \\
\text{s.t.} & \underbrace{X^{(v)} M^{(v)} F^{(v)} = U^{(v)}}_{\text{Refined anchor refactoring}}, S = \Psi \left(\underbrace{\left[B^{(1)}; \dots; B^{(V)}; P \right]}_{\text{Unified tensorized}} \right), \\
& \omega^{(v)} \geq 0, \|\omega\|_2^2 = 1, y_{ij} \in \{0, 1\}, M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}, \\
& F^{(v)T} F^{(v)} = I_m, G^{(v)T} G^{(v)} = I_m, P P^T = I_m, H^T H = I_c, \\
& \Phi = \{F^{(v)}, U^{(v)}, B^{(v)}, G^{(v)}, P, H, Y, \omega, Q\},
\end{aligned} \tag{11}$$

where Φ denotes the set of variables to be jointly optimized.

4. Model alternate optimization

This section elaborates the optimization process for the proposed model. The optimization problem is solved within an Augmented Lagrangian Multiplier (ALM) framework, combined with an Alternating Direction Minimization (ADM) strategy. To solve the proposed model efficiently, we introduce an auxiliary variable $Q = S$ and formulate the augmented Lagrangian function as follows:

$$\begin{aligned}
\mathcal{L}(F^{(v)}, U^{(v)}, B^{(v)}, G^{(v)}, P, H, Y, \omega, Q) \\
= \sum_{v=1}^V \left\{ \left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2 + \langle K^{(v)}, U^{(v)} - X^{(v)} M^{(v)} F^{(v)} \rangle \right. \\
+ \frac{\mu}{2} \left\| U^{(v)} - X^{(v)} M^{(v)} F^{(v)} \right\|_F^2 \left. \right\} - \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] \\
+ \beta \sum_{i=1}^c \sum_{j=1}^n y_{ij} \left\| H_{(:,i)} - P_{(:,j)} \right\|_2^2 + \gamma \|Q\|_{\pi, \Theta} \\
+ \langle R, Q - S \rangle + \frac{\rho}{2} \|Q - S\|_F^2 \\
\text{s.t. } S = \Psi \left(\left[B^{(1)}; \dots; B^{(V)}; P \right] \right), \\
\omega^{(v)} \geq 0, \|\omega\|_2^2 = 1, y_{ij} \in \{0, 1\}, M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}, \\
F^{(v)T} F^{(v)} = I_m, G^{(v)T} G^{(v)} = I_m, P P^T = I_m, H^T H = I_c.
\end{aligned} \tag{12}$$

Optimization variables $F^{(v)}$: Since $F^{(v)}$ is subject to the orthogonality constraint, we employ a two-step approach to solve it. The first step is solving the subproblem without OC via:

$$\min_{F^{(v)}} \sum_{v=1}^V \frac{\mu}{2} \left\| U^{(v)} - X^{(v)} M^{(v)} F^{(v)} + \frac{K^{(v)}}{\mu} \right\|_F^2 \tag{13}$$

Solving for each $F^{(v)}$ individually, delete irrelevant items and taking the derivative to zero. So we can obtain the following intermediate solution:

$$\widehat{F}^{(v)} = (X^{(v)} M^{(v)})^{-1} T_1^{(v)}, \tag{14}$$

where $T_1^{(v)} = U^{(v)} + \frac{K^{(v)}}{\mu}$. The next step is imposing OC: $F^{(v)T} F^{(v)} = I_m$. Then, we perform SVD on $\widehat{F}^{(v)}$: $\widehat{F}^{(v)} = U_1 \Sigma_1 V_1^T$. The optimal solution for $F^{(v)}$ with orthogonality is given by $F^{(v)*} = U_1 V_1^T$. The two-step procedure yields $F^{(v)}$ with reduced computational cost.

Optimization variables $U^{(v)}$: By fixing the other variables, the variable $U^{(v)}$ is updated by solving the following subproblem:

$$\begin{aligned}
\min_{U^{(v)}} & \sum_{v=1}^V \left\{ \left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2 \right. \\
& \left. + \frac{\mu}{2} \left\| U^{(v)} - X^{(v)} M^{(v)} F^{(v)} + \frac{K^{(v)}}{\mu} \right\|_F^2 \right\}
\end{aligned} \tag{15}$$

Solving for each $U^{(v)}$ individually, we can obtain:

$$U^{(v)} = T_2^{(v)} \left(T_3^{(v)} \right)^{-1}, \tag{16}$$

where $T_2^{(v)} = 2X^{(v)} B^{(v)T} + \mu X^{(v)} M^{(v)} F^{(v)} - K^{(v)}$ and $T_3^{(v)} = 2B^{(v)} M^{(v)} M^{(v)T} B^{(v)T} + \mu I_m$. Let $\widehat{B}^{(v)} = B^{(v)} M^{(v)}$ and $\widehat{X}^{(v)} = X^{(v)} M^{(v)}$ represent the subsets of corresponding existing elements, the solution of $U^{(v)}$ can be characterized by:

$$U^{(v)*} = T_4^{(v)} \left(T_5^{(v)} \right)^{-1}, \tag{17}$$

where $T_4^{(v)} = 2X^{(v)} B^{(v)T} + \mu \widehat{X}^{(v)} F^{(v)} - K^{(v)}$ and $T_5^{(v)} = 2\widehat{B}^{(v)} \widehat{B}^{(v)T} + \mu I_m$. **Optimization variables $B^{(v)}$:** By fixing the other variables, the variable $B^{(v)}$ is updated by solving the following subproblem:

$$\begin{aligned}
\min_{B^{(v)} T B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}} & \left\{ \sum_{v=1}^V \left\| U^{(v)} B^{(v)} M^{(v)} - X^{(v)} M^{(v)} \right\|_F^2 \right. \\
& \left. - \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] + \frac{\rho}{2} \left\| Q - S + \frac{R}{\rho} \right\|_F^2 \right\}
\end{aligned} \tag{18}$$

Since $B^{(v)}$ is restricted by $M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}$, it is solved through two-step process, this manner similar to $F^{(v)}$. Solving for each $B^{(v)}$ individually and not considering OC currently, we further have:

$$\min_{B^{(v)}} \text{Tr} \left[B^{(v)T} \left(U^{(v)T} T_6^{(v)} M^{(v)} M^{(v)T} + T_7^{(v)} \right) - \alpha \omega^{(v)} T_8^{(v)} \right], \tag{19}$$

where $T_6^{(v)} = U^{(v)} B^{(v)} - 2X^{(v)}$, $T_7^{(v)} = \frac{\rho}{2} B^{(v)} - \rho Q^{(v)} - R^{(v)}$ and $T_8^{(v)} = B^{(v)T} P^T G^{(v)}$. Taking the derivative with respect to $B^{(v)}$ and setting it to zero, which can be converted to:

$$2U^{(v)T} T_9^{(v)} - \alpha \omega^{(v)} G^{(v)} P + T_{10}^{(v)} = 0, \tag{20}$$

where $T_9^{(v)} = U^{(v)} B^{(v)} M^{(v)} M^{(v)T} - X^{(v)}$ and $T_{10}^{(v)} = \rho(B^{(v)} - Q^{(v)}) - R^{(v)}$. The conventional method uses Sylvester equation (Sorensen & Antoulas, 2002) to solve the above equation. However, since Sylvester equation has complexity $\mathcal{O}(n^3)$, we use approximation to reduce computational complexity. By replacing $B^{(v)} M^{(v)} M^{(v)T}$ with $B^{(v)}$, we can obtain:

$$\widehat{B}^{(v)} = \left(T_{11}^{(v)} \right)^{-1} T_{12}^{(v)}, \tag{21}$$

where $T_{11}^{(v)} = 2U^{(v)T} U^{(v)} + \rho I_m$ and $T_{12}^{(v)} = 2U^{(v)T} X^{(v)} + \alpha \omega^{(v)} G^{(v)} P + \rho Q^{(v)} + R^{(v)}$. The next step is imposing OC: $M^{(v)T} B^{(v)T} B^{(v)} M^{(v)} = I_{n_v}$, we perform SVD on $\widehat{B}^{(v)}$: $\widehat{B}^{(v)} = U_2 \Sigma_2 V_2^T$. The optimal solution for $B^{(v)}$ is then given by:

$$B^{(v)*} = U_2 V_2^T \odot W^{(v)}, \tag{22}$$

where $W^{(v)} = \mathbf{1}_{m \times 1} \cdot [\sum_{r=1}^{n_v} M^{(v)}(1, r), \dots, \sum_{r=1}^{n_v} M^{(v)}(n, r)]_{1 \times n} \in \mathbb{R}^{m \times n}$. Hence, optimizing $B^{(v)}$ takes $\mathcal{O}(nmd_v + nm^2)$.

Optimization variables $G^{(v)}$: The update rule for the alignment matrix $G^{(v)}$ is derived by solving the following constrained subproblem:

$$\max_{G^{(v)T} G^{(v)} = I_m} \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] \tag{23}$$

By solving for each $G^{(v)}$ individually, this problem can be formulated as:

$$\max_{G^{(v)T} G^{(v)} = I_m} \text{Tr} \left[G^{(v)T} \left(\omega^{(v)} P B^{(v)T} \right) \right] \tag{24}$$

Let $\zeta^{(v)} = P B^{(v)T}$ and compute its SVD: $\zeta^{(v)} = U_3 \Sigma_3 V_3^T$. The optimal solution for $G^{(v)}$ that maximizes the trace under OC is then given

by: $G^{(v)*} = U_3 V_3^T$. This solution ensures that $G^{(v)}$ is orthogonal matrix ($G^{(v)T} G^{(v)} = I_m$) while maximizing the alignment between the bipartite graph structures and the consistent representation.

Optimization variables P : Fixing all other variables, the optimization problem with respect to P can be reduced to the follows:

$$\min_{P \in \mathbb{R}^{m \times n}} \left\{ -\alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] + \beta \sum_{i=1}^c \sum_{j=1}^n y_{ij} \|H_{(:,i)} - P_{(:,j)}\|_2^2 + \frac{\rho}{2} \|Q^{(V+1)} - P + \frac{R^{(V+1)}}{\rho}\|_F^2 \right\} \quad (25)$$

Defining the diagonal matrices: $D_n = \text{diag}(Y^T \mathbf{1}_{c \times 1}) = I_n$, $D_c = \text{diag}(Y \mathbf{1}_{n \times 1}) = Y Y^T$, the second term of Eq. (25) can be derived to:

$$\begin{aligned} & \sum_{i=1}^c \sum_{j=1}^n y_{ij} \|H_{(:,i)} - P_{(:,j)}\|_2^2 \\ &= \sum_{i=1}^c \sum_{j=1}^n y_{ij} (H_{(:,i)} - P_{(:,j)})^T (H_{(:,i)} - P_{(:,j)}) \\ &= \text{Tr} (P D_n P^T - 2 P Y^T H^T + H D_c H^T) \\ &= \text{Tr} (P^T P - 2 H Y P^T + Y^T H^T H Y) \end{aligned} \quad (26)$$

Substituting back into the original objective, Eq. (25) converts to a maximization problem as:

$$\max_{P \in \mathbb{R}^{m \times n}} \text{Tr} \left[P^T \left(\alpha \sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} + T_{13} \right) \right], \quad (27)$$

where $T_{13} = 2\beta H Y + \rho Q^{(V+1)} + R^{(V+1)}$. Defining the combined matrix: $\Lambda = \alpha \sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} + T_{13}$. Let the SVD of Λ be: $\Lambda = U_4 \Sigma_4 V_4^T$, the optimal solution for P under the OC is then: $P^* = U_4 V_4^T$.

Optimization variables H : By fixing the other variables, the variable H is updated by solving the following subproblem:

$$\min_{H^T H = I_c} \sum_{i=1}^c \sum_{j=1}^n y_{ij} \|H_{(:,i)} - P_{(:,j)}\|_2^2 \quad (28)$$

The Eq. (28) can be rewritten as trace form:

$$\min_{H^T H = I_c} \text{Tr} [P^T P - 2 P Y^T H^T + Y^T H^T H Y] \quad (29)$$

By discarding constant terms and noting that $H^T H = I_c$, the problem simplifies to:

$$\max_{H^T H = I_c} \text{Tr} [H^T (P Y^T)] \quad (30)$$

Let the SVD of $P Y^T$ be $U_5 \Sigma_5 V_5^T$. Then the optimal solution for H is given by: $H^* = U_5 V_5^T$.

Optimization variables Y : By fixing the other variables, the variable Y is updated via the solution of follows:

$$\min_{Y \in \{0,1\}} \sum_{i=1}^c \sum_{j=1}^n y_{ij} \|H_{(:,i)} - P_{(:,j)}\|_2^2, \quad (31)$$

where $H \in \mathbb{R}^{m \times c}$, $P \in \mathbb{R}^{m \times n}$ and $Y \in \{0,1\}^{c \times n}$ is the label matrix. The update rule for Y is given by:

$$Y_{i,j} = \begin{cases} 1, & \text{if } \arg \min_{i^*} \|H_{(:,i^*)} - P_{(:,j)}\|_2^2 = i, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

Optimization variables $\omega^{(v)}$: By fixing the other variables, the variable $\omega^{(v)}$ is updated by solving the following subproblem:

$$\max_{\omega^{(v)} \geq 0, \|\omega\|_2^2 = 1} \alpha \text{Tr} \left[\left(\sum_{v=1}^V \omega^{(v)} G^{(v)} B^{(v)} \right) P^T \right] \quad (33)$$

Let $t^{(v)} = \text{Tr} (G^{(v)} B^{(v)} P^T)$ and $t = (t^{(1)}, t^{(2)}, \dots, t^{(V)})$, by the Cauchy-Schwarz inequality (Tang et al., 2023), the closed-form solution as follows:

$$\omega^{(v)*} = \frac{|t^{(v)}|}{\|t\|_2}, \quad (34)$$

where $\|t\|_2 = \sqrt{\sum_{v=1}^V (t^{(v)})^2}$ and $|t^{(v)}|$ presents the absolute value of $t^{(v)}$. **Optimization Tensor Q :** With other variables keeping fixed, the auxiliary tensor Q is updated by solving the following subproblem:

$$\min_Q \gamma \|Q\|_{\pi, \Theta} + \frac{\rho}{2} \left\| Q - \left(S + \frac{R}{\rho} \right) \right\|_F^2 \quad (35)$$

Eq. (35) is equivalent to solving:

$$Q^* = \arg \min_Q \frac{\gamma}{\rho} \|Q\|_{\pi, \Theta} + \frac{1}{2} \|Q - H\|_F^2, \quad (36)$$

where $H = S + (R/\rho)$. Eq. (36) represents a problem related to the WTNN norm, which can be addressed by Theorem 1.

Theorem 1. The solution of Eq. (36) can be obtained through the application of the tensor tubal-weighted shrinkage operator:

$$Q^* = \mathcal{U} * C_{\tau \times \pi}(\mathcal{O}) * \mathcal{V}^T, \quad (37)$$

where $\tau = \gamma/\rho > 0$, t -SVD operation $\mathcal{H} = \mathcal{U} * \mathcal{O} * \mathcal{V}^T$, and $C_{\tau \times \pi}(\mathcal{O}) = \mathcal{O} * \mathcal{D}$. Moreover, $\mathcal{D}$ denotes a diagonal tensor whose diagonal elements in the Fourier domain are given by

$$D_f(a, a, b) = \left[1 - \frac{\tau \times \pi}{\mathcal{O}_f^{(b)}(a, a)} \right]_+ \quad (38)$$

Updating the Lagrangian multipliers and penalty factors: By fixing all other variables, the Lagrange multipliers and penalty factors can be updated according to the following rules:

$$\begin{cases} \mathcal{R} \leftarrow \mathcal{R} + \rho(Q - S), \\ \mathcal{K}^{(v)} \leftarrow \mathcal{K}^{(v)} + \mu(U^{(v)} - X^{(v)} M^{(v)} F^{(v)}), \\ \rho^* \leftarrow \min \{ \lambda_1 \rho, \rho_{\max} \}, \\ \mu^* \leftarrow \min \{ \lambda_2 \mu, \mu_{\max} \}, \end{cases} \quad (39)$$

where λ_1 and λ_2 are auto-adjusting parameters used to update ρ and μ toward them maximum value.

The iterative optimization process terminates when the following convergence condition is satisfied:

$$\|Q - S\|_{\infty} < \epsilon \quad (40)$$

To streamline the presentation, the entire optimization procedure is summarized in Algorithm 1.

5. Complexity analysis

5.1. Time complexity

The analysis of computational complexity highlights the efficiency and scalability of the proposed algorithm, which depends on the collaborative operation of several solvers represented by $F^{(v)}$, $U^{(v)}$, $B^{(v)}$, $G^{(v)}$, $\omega^{(v)}$, P , H , Y , Q . Specifically, for $F^{(v)}$, its computational complexity is $\mathcal{O}(n_v m d_v)$. When updating $U^{(v)}$, its complexity is $\mathcal{O}(n m d_v)$. When analytically obtaining $B^{(v)}$, it costs $\mathcal{O}(n m d_v)$ for all columns. When updating $G^{(v)}$ and $\omega^{(v)}$ need $\mathcal{O}(n m^2)$. The computational cost of calculating P and H is $\mathcal{O}(n m c)$. To obtain an optimal Y , it needs $\mathcal{O}(n c)$. For tensor Q , each updating involves FFT, inverse FFT, and t -SVD operations, with complexity $\mathcal{O}(2 n m (V+1) \log(n) + n m (V+1)^2)$. Here, define the number of iterations as $iter$ (with $iter \ll n$). The total time complexity is $Total_{time} = \mathcal{O}[iter \times (n m (d + (V+1) \log(n)))]$, where $d = \sum_{v=1}^V d_v$. Therefore, the time complexity can be approximated as $\mathcal{O}(n \log(n))$.

5.2. Space complexity

The dominant space usage of SIMC-ARTB arises from matrices $F^{(v)} \in \mathbb{R}^{n_v \times m}$, $U^{(v)} \in \mathbb{R}^{d_v \times m}$, $B^{(v)} \in \mathbb{R}^{m \times n}$, $G^{(v)} \in \mathbb{R}^{m \times m}$, $\omega \in \mathbb{R}^{V \times 1}$, $K^{(v)} \in \mathbb{R}^{d_v \times m}$, $P \in \mathbb{R}^{m \times n}$, $H \in \mathbb{R}^{m \times c}$, $Y \in \mathbb{R}^{c \times n}$, and $Q, S, R \in \mathbb{R}^{m \times n \times (V+1)}$. Consequently, the overall space complexity of SIMC-ARTB can be expressed

Table 2
Description of all test datasets.

No.	Datasets	Category	Samples n	Views V	Clusters c	Dimensions d_v
D1	NGs	News	500	3	5	[2000, 2000, 2000]
D2	Flowers17	Flower	1360	7	17	[1360, 1360, 1360, 1360, 1360, 1360, 1360]
D3	100leaves	Plant	1600	3	100	[64, 64, 64]
D4	CCV	Event	6773	3	20	[20, 20, 20]
D5	Caltech101	Object	9144	5	102	[48, 40, 254, 512, 928]
D6	SUNRGBD	Scene	10,335	2	45	[4096, 4096]
D7	Animal	Animal	11,673	4	20	[2689, 2000, 2001, 2000]
D8	NUSWIDE OBJ	Object	30,000	5	31	[65, 226, 145, 74, 129]

Algorithm 1 The pipeline for optimization SIMC-ARTB.

Input: The incomplete multi-view samples $\{X^{(v)}\}_{v=1}^V$, the incomplete indicator $M^{(v)}$, the trade-off parameters α , β and γ , the cluster number c , and the anchors number m .

```

1: Initialization:  $B^{(v)} = \mathbf{1}$ ,  $F^{(v)} = \mathbf{1}$ ,  $U^{(v)} = \mathbf{1}$ ;  $K^{(v)} = \mathbf{0}$ ,  $R^{(v)} = \mathbf{0}$ ,  $Q^{(v)} = \mathbf{0}$ ,  $P = \mathbf{0}$ ,  $Y = \mathbf{0}$ ;  $G^{(v)} = I$ ,  $H = I$ ;  $\rho = 10^{-4}$ ,  $\rho_{\max} = 10^{10}$ ;  $\mu = 10^{-5}$ ,  $\mu_{\max} = 10^{10}$ ;  $\lambda_1 = 1.9$ ,  $\lambda_2 = 1.5$ ;  $\omega^{(v)} = 1/V$ ,  $m = c$ ;  $iter = 0$ ,  $iter_{\max} = 40$ ,  $\epsilon = 10^{-5}$ .
2: while not converged do
3:    $iter = iter + 1$ ;
4:   for  $v = 1$  to  $V$  do
5:     Update  $F^{(v)}$  by solving Eq. (14);
6:     Update  $U^{(v)}$  by solving Eq. (17);
7:     Update  $B^{(v)}$  by solving Eq. (22);
8:     Update  $G^{(v)}$  by solving Eq. (24);
9:     Update  $\omega^{(v)}$  by solving Eq. (34);
10:  end for
11:  Update  $P$  by solving Eq. (27);
12:  Update  $H$  by solving Eq. (30);
13:  Update  $Y$  by solving Eq. (32);
14:  Update  $Q$  by solving Eq. (36);
15:  Update the Lagrangian multipliers and penalty factors using Eq. (39);
16:  Check the following convergence conditions:
17:  Using Eq. (40) and  $iter < iter_{\max}$ ;
18: end while

```

Output: The clustering label matrix Y .

as $Total_{space} = \mathcal{O}[nm(V+1) + nd_{\max}]$, $d_{\max} = \max(d_v)$, which grows linearly with the data size n .

6. Experiments

In this section, we conduct a series of carefully designed experiments to systematically assess multiple aspects of the proposed SIMC-ARTB framework, including its clustering performance, parameter sensitivity, ablation analysis, visualization of weights and block-diagonal structures, anchor number study, execution time comparison, and convergence behavior. All experiments were performed using MATLAB R2025a on a Windows 11 (x64) platform equipped with an Intel Core U7-265K processor (3.90 GHz) and 64GB of RAM.

6.1. Experimental settings

6.1.1. Datasets description

Our proposed method, along with the compared approaches, is evaluated on eight publicly available benchmark datasets, namely: NGs¹,

Flowers17², 100leaves³, CCV⁴, Caltech101⁵, SUNRGBD⁶, Animal (Xu et al., 2024), and NUSWIDE OBJ (Chua et al., 2009). A summary of the key characteristics of these benchmark datasets is provided in Table 2.

- 1) **NGs** comprises 500 newsgroup documents across five categories, and each document undergoes three distinct pre-processing methods.
- 2) **Flowers17** includes 1360 images organized into 17 flower categories (80 samples per class). Each image is described through seven distinct visual feature views.
- 3) **100leaves** consists of 1600 samples representing 100 distinct plant species, with each sample captured from three different views.
- 4) **CCV** contains 6773 samples drawn from 20 semantic categories of YouTube videos, where each sample is characterized by three views, all sharing the same feature dimensionality of 20.
- 5) **Caltech101** comprises of 9144 image samples belonging to 102 distinct clusters. Each object represented by five types of features to simulate multi-view observations.
- 6) **SUNRGBD** includes 10,335 real RGB-D images depicting indoor scenes, categorized into 45 clusters, with each RGB image associated with both a depth map and a segmentation map.
- 7) **Animal** consists of 11,673 animal images that can be divided into 20 mutually exclusive classes. Four commonly used image features are extracted to construct a four-view representation for each sample.
- 8) **NUSWIDE OBJ** contains 30,000 web samples that are from 5 views and 31 classes. Each sample is characterized by five feature views with dimensionalities of 65, 226, 145, 74, and 129, respectively.

For the datasets described above, incomplete multi-view datasets are generated by randomly removing a proportion of samples from each view, where the missing rate (MR) is selected from [10%, 30%, 50%]. To ensure that every sample remains observable, any instance that becomes missing in all views is randomly assigned to retain its observation in one view. In our experiments, a single random missing-view mask is generated for each missing ratio and is subsequently fixed for all methods. For instance, the original NGs dataset is transformed into a missing-view dataset with $MR = 10\%$, denoted by NGs_10%, and all subsequent experiments are performed on this fixed dataset without generating additional missing-view masks.

6.1.2. Compared methods and parameters

To validate the effectiveness and superiority of the proposed SIMC-ARTB, we conduct comparisons with several representative algorithms.

- 1) **SIMVC-SA** (ACMM'23) (Wen et al., 2023): It presents a scalable framework that jointly optimizes incomplete anchor graph construction and structure alignment across multiple views to achieve efficient and accurate multi-view clustering.

² <http://www.robots.ox.ac.uk/~vgg/data/flowers/17/>.

³ <https://archive.ics.uci.edu/dataset/241/one+hundred+plant+species+leaves+data+set>.

⁴ <https://www.ee.columbia.edu/~ln/dvmm/CCV/>.

⁵ <https://www.vision.caltech.edu/datasets/>.

⁶ <https://rgbd.cs.princeton.edu/>.

¹ <https://lig-membres.imag.fr/grimal/data.html>.

Table 3The experiment results [*means(std)*] on eight incomplete datasets with different *MR*.

Datasets	Methods/MR	ACC (% ↑)			NMI (% ↑)			PUR (% ↑)		
		10%	30%	50%	10%	30%	50%	10%	30%	50%
NGs	SIMVC-SA('23)	96.60(0.00)	96.40(0.00)	96.60(0.00)	89.52(0.00)	89.11(0.00)	89.52(0.00)	96.60(0.00)	96.40(0.00)	96.60(0.00)
	CCA-IMVC('23)	96.20(0.00)	92.06(0.00)	83.40(0.00)	88.51(0.00)	78.85(0.00)	63.95(0.00)	96.20(0.00)	92.06(0.00)	83.40(0.00)
	SCBGL('23)	97.20(0.00)	97.40(0.00)	97.20(0.00)	91.54(0.00)	91.71(0.00)	91.48(0.00)	97.20(0.00)	97.40(0.00)	97.20(0.00)
	DAQINT('24)	95.40(0.00)	90.80(0.00)	77.98(0.00)	86.78(0.00)	75.90(0.00)	57.63(0.00)	95.40(0.00)	90.80(0.00)	77.98(0.00)
	DCGA('25)	97.16(0.00)	93.40(0.00)	81.10(0.00)	91.06(0.08)	81.87(0.00)	58.22(0.00)	97.16(0.00)	93.40(0.00)	81.10(0.00)
	OIMC-IAGD('26)	71.70(8.87)	60.12(10.55)	37.58(6.78)	55.30(7.73)	37.44(6.42)	15.34(7.38)	72.62(7.82)	61.74(8.64)	39.26(6.77)
	PBDAL('25)	99.40(0.00)	99.14(0.00)	99.16(0.00)	97.91(0.00)	97.11(0.00)	97.23(0.00)	99.40(0.00)	99.14(0.00)	99.16(0.00)
	ASCR('25)	99.99(0.01)	98.35(1.31)	99.73(0.02)	99.99(0.02)	99.19(0.64)	99.06(0.06)	99.99(0.01)	98.80(0.92)	99.73(0.02)
Flowers17	Ours	<u>99.80(0.00)</u>	99.60(0.00)	<u>99.20(0.00)</u>	<u>99.30(0.00)</u>	<u>98.61(0.00)</u>	<u>97.21(0.00)</u>	<u>99.80(0.00)</u>	99.60(0.00)	<u>99.20(0.00)</u>
	SIMVC-SA('23)	52.82(0.00)	53.63(0.00)	51.32(0.00)	52.01(0.00)	51.64(0.00)	49.86(0.00)	54.04(0.00)	54.39(0.00)	51.98(0.00)
	CCA-IMVC('23)	51.16(0.00)	45.99(0.00)	42.30(0.00)	49.57(0.00)	43.82(0.00)	39.10(0.00)	52.14(0.00)	48.04(0.00)	43.22(0.00)
	SCBGL('23)	47.47(0.00)	47.85(0.00)	47.91(0.00)	47.45(0.00)	47.65(0.00)	47.47(0.00)	48.65(0.00)	49.22(0.00)	49.22(0.00)
	DAQINT('24)	44.93(0.00)	30.30(0.00)	26.83(0.00)	43.14(0.00)	32.83(0.00)	27.33(0.00)	46.29(0.00)	31.99(0.00)	28.29(0.00)
	DCGA('25)	40.35(0.04)	37.58(0.00)	32.45(0.00)	42.33(0.05)	39.07(0.00)	33.43(0.00)	41.81(0.14)	39.80(0.00)	34.03(0.00)
	OIMC-IAGD('26)	49.49(1.34)	43.65(1.56)	37.73(2.08)	50.44(0.68)	45.10(1.23)	37.09(1.72)	50.92(1.09)	45.88(2.05)	38.96(2.15)
	PBDAL('25)	90.68(0.00)	63.29(0.00)	39.24(0.00)	90.55(0.00)	69.76(0.00)	43.01(0.00)	91.17(0.00)	64.72(0.00)	41.18(0.00)
100leaves	ASCR('25)	72.78(1.21)	74.50(0.81)	74.62(1.20)	83.20(0.55)	84.08(0.24)	84.42(0.50)	76.10(0.91)	77.46(0.59)	77.36(0.98)
	Ours	92.87(0.00)	91.54(0.00)	90.81(0.00)	92.48(0.00)	91.63(0.00)	90.77(0.00)	92.87(0.00)	91.54(0.00)	90.81(0.00)
	SIMVC-SA('23)	71.48(0.00)	71.86(0.00)	71.15(0.00)	86.38(0.00)	86.46(0.00)	86.13(0.00)	74.16(0.00)	74.44(0.00)	73.66(0.00)
	CCA-IMVC('23)	56.61(0.00)	40.62(0.00)	31.64(0.00)	76.59(0.00)	65.95(0.00)	60.91(0.00)	59.33(0.00)	42.74(0.00)	33.88(0.00)
	SCBGL('23)	70.34(0.00)	69.23(0.00)	69.96(0.00)	85.50(0.00)	84.99(0.00)	85.57(0.00)	72.75(0.00)	71.76(0.00)	73.00(0.00)
	DAQINT('24)	59.83(0.00)	42.18(0.00)	34.08(0.00)	78.85(0.00)	67.68(0.00)	62.91(0.00)	62.24(0.00)	44.73(0.00)	36.77(0.00)
	DCGA('25)	61.83(0.08)	41.48(0.00)	31.91(0.00)	80.55(0.03)	66.93(0.00)	60.99(0.00)	64.65(0.09)	43.98(0.00)	34.16(0.00)
	OIMC-IAGD('26)	70.17(1.93)	56.37(2.02)	41.48(1.16)	86.75(0.70)	76.66(0.71)	67.98(0.44)	73.31(1.67)	58.97(1.74)	44.25(0.85)
CCV	PBDAL('25)	86.74(0.00)	86.74(0.00)	84.16(0.00)	95.80(0.00)	95.99(0.00)	94.13(0.00)	89.46(0.00)	89.33(0.00)	86.69(0.00)
	ASCR('25)	81.22(0.52)	80.33(0.56)	80.17(0.56)	93.90(0.13)	95.10(0.11)	93.94(0.13)	84.17(0.38)	84.25(0.44)	83.72(0.42)
	Ours	96.31(0.00)	93.88(0.00)	88.50(0.00)	98.48(0.00)	96.41(0.00)	94.26(0.00)	96.88(0.00)	93.88(0.00)	88.88(0.00)
	SIMVC-SA('23)	22.32(0.00)	22.80(0.00)	22.28(0.00)	19.11(0.00)	18.66(0.00)	18.74(0.00)	25.27(0.00)	26.11(0.00)	25.63(0.00)
	CCA-IMVC('23)	21.42(0.00)	19.37(0.00)	17.57(0.00)	17.17(0.00)	14.51(0.00)	12.87(0.00)	24.68(0.00)	22.28(0.00)	21.00(0.00)
	SCBGL('23)	23.69(0.00)	23.74(0.00)	23.47(0.00)	18.53(0.00)	18.78(0.00)	18.61(0.00)	26.93(0.00)	27.23(0.00)	26.90(0.00)
	DAQINT('24)	18.94(0.00)	15.89(0.00)	14.95(0.00)	15.07(0.00)	12.14(0.00)	10.54(0.00)	22.75(0.00)	19.64(0.00)	18.20(0.00)
	DCGA('25)	17.22(0.04)	15.36(0.00)	13.39(0.00)	15.25(0.07)	12.84(0.00)	11.00(0.00)	21.83(0.03)	20.03(0.00)	17.74(0.00)
Caltech101	OIMC-IAGD('26)	13.59(2.86)	15.23(0.65)	14.10(0.43)	9.58(3.37)	10.46(0.39)	9.66(0.25)	16.79(2.56)	18.95(0.60)	17.57(0.32)
	PBDAL('25)	19.30(0.00)	17.51(0.00)	17.43(0.00)	15.88(0.00)	13.41(0.00)	12.28(0.00)	22.77(0.00)	21.91(0.00)	21.05(0.00)
	ASCR('25)	65.67(1.78)	60.90(1.55)	63.60(1.19)	73.73(0.60)	70.74(0.48)	67.37(0.46)	71.64(1.11)	66.43(0.96)	67.81(0.94)
	Ours	76.23(0.00)	70.06(0.00)	67.03(0.00)	74.33(0.00)	72.87(0.00)	67.68(0.00)	78.09(0.00)	71.65(0.00)	70.35(0.00)
	SIMVC-SA('23)	25.57(0.00)	25.72(0.00)	25.70(0.00)	48.50(0.00)	48.45(0.00)	48.43(0.00)	46.83(0.00)	46.54(0.00)	46.54(0.00)
	CCA-IMVC('23)	25.85(0.00)	23.27(0.00)	19.18(0.00)	47.68(0.00)	43.22(0.00)	36.67(0.00)	46.04(0.00)	41.64(0.00)	34.52(0.00)
	SCBGL('23)	27.53(0.00)	27.23(0.00)	27.27(0.00)	49.40(0.00)	49.30(0.00)	49.25(0.00)	47.93(0.00)	47.50(0.00)	47.60(0.00)
	DAQINT('24)	23.41(0.00)	19.51(0.00)	15.68(0.00)	40.45(0.00)	39.79(0.00)	34.34(0.00)	38.26(0.00)	38.09(0.00)	31.71(0.00)
SUNRGBD	DCGA('25)	19.92(0.02)	18.26(0.00)	15.84(0.00)	41.41(0.03)	38.19(0.00)	33.61(0.00)	39.31(0.04)	36.24(0.00)	31.97(0.00)
	OIMC-IAGD('26)	27.02(0.73)	25.43(1.16)	22.35(0.92)	43.50(0.22)	41.38(0.44)	36.82(0.33)	41.65(0.17)	39.55(0.38)	35.56(0.27)
	PBDAL('25)	34.14(0.00)	23.76(0.00)	19.61(0.00)	57.40(0.00)	44.18(0.00)	37.51(0.00)	55.91(0.00)	41.93(0.00)	35.78(0.00)
	ASCR('25)	56.45(0.52)	55.14(0.48)	56.61(0.59)	85.67(0.16)	83.42(0.17)	86.17(0.19)	79.93(0.31)	79.64(0.35)	81.22(0.36)
	Ours	66.25(0.00)	64.95(0.00)	60.23(0.00)	86.05(0.00)	84.77(0.00)	82.99(0.00)	87.26(0.00)	84.63(0.00)	82.15(0.00)
	SIMVC-SA('23)	19.61(0.00)	19.51(0.00)	19.57(0.00)	27.46(0.00)	27.36(0.00)	27.41(0.00)	40.54(0.00)	40.25(0.00)	40.39(0.00)
	CCA-IMVC('23)	18.54(0.00)	17.02(0.00)	14.30(0.00)	25.20(0.00)	16.85(0.00)	16.11(0.00)	38.42(0.00)	28.78(0.00)	27.47(0.00)
	SCBGL('23)	18.92(0.00)	19.73(0.00)	20.11(0.00)	26.28(0.00)	20.10(0.00)	22.95(0.00)	39.21(0.00)	30.64(0.00)	35.17(0.00)
Animal	DAQINT('24)	19.05(0.00)	16.56(0.00)	14.89(0.00)	23.00(0.00)	20.31(0.00)	17.45(0.00)	35.66(0.00)	32.27(0.00)	29.31(0.00)
	DCGA('25)	17.93(0.08)	15.31(0.00)	13.77(0.00)	25.69(0.01)	21.88(0.00)	18.83(0.00)	37.91(0.05)	33.17(0.00)	30.50(0.00)
	OIMC-IAGD('26)	21.54(0.95)	17.28(0.72)	15.26(0.43)	24.84(0.27)	20.29(0.32)	16.32(0.40)	37.76(0.58)	29.16(0.61)	23.89(0.47)
	PBDAL('25)	19.43(0.00)	16.13(0.00)	13.14(0.00)	25.52(0.00)	21.15(0.00)	16.72(0.00)	37.68(0.00)	33.12(0.00)	29.44(0.00)
	ASCR('25)	22.58(0.17)	21.78(0.26)	22.34(0.18)	35.51(0.10)	36.74(0.15)	37.89(0.13)	46.46(0.16)	42.32(0.20)	45.50(0.17)
	Ours	27.05(0.00)	26.39(0.00)	23.62(0.00)	33.32(0.00)	31.99(0.00)	27.05(0.00)	45.72(0.00)	44.78(0.00)	41.49(0.00)
	SIMVC-SA('23)	15.83(0.00)	16.23(0.00)	15.99(0.00)	11.56(0.00)	13.85(0.00)	14.48(0.00)	17.02(0.00)	20.36(0.00)	19.77(0.00)
	CCA-IMVC('23)	16.41(0.00)	16.12(0.00)	15.40(0.00)	12.46(0.00)	11.34(0.00)	10.41(0.00)	19.71(0.00)	19.30(0.00)	18.03(0.00)
NUSWIDE0BJ	SCBGL('23)	17.20(0.00)	17.31(0.00)	17.75(0.00)	12.50(0.00)	13.61(0.00)	12.55(0.00)	19.40(0.00)	20.77(0.00)	20.20(0.00)
	DAQINT('24)	15.27(0.00)	13.09(0.00)	11.87(0.00)	11.12(0.00)	8.68(0.00)	7.41(0.00)	18.34(0.00)	16.21(0.00)	15.12(0.00)
	DCGA('25)	19.26(0.02)	18.06(0.00)	15.48(0.00)	16.79(0.03)	15.45(0.00)	12.24(0.00)	23.08(0.07)	22.04(0.00)	19.04(0.00)
	OIMC-IAGD('26)	17.52(0.52)	15.61(0.44)	13.58(0.35)	13.77(0.42)	11.63(0.38)	9.47(0.11)	20.84(0.56)	18.95(0.48)	16.71(0.38)
	PBDAL('25)	16.11(0.00)	12.78(0.00)	12.40(0.00)	12.70(0.00)	7.86(0.00)	8.50(0.00)	19.79(0.00)	15.50(0.00)	17.43(0.00)
	ASCR('25)	80.32(1.16)	81.44(1.95)	79.38(1.10)	88.08(0.48)	89.88(0.57)	87.90(0.39)	86.73(0.85)	86.39(1.32)	84.58(1.00)
	Ours	89.42(0.00)	87.54(0.00)	84.61(0.00)	88.35(0.00)	87.63(0.00)	87.02(0.00)	91.80(0.00)	90.13(0.00)	87.57(0.00)
	SIMVC-SA('23)	15.21(0.00)	15.30(0.00)	15.74(0.00)	4.18(0.00)	8.52(0.00)	4.72(0.00)	15.40(0.00)	18.70(0.00)	16.30(0.00)
NUSWIDE0BJ	CCA-IMVC('23)	13.66(0.00)	13.02(0.00)	12.22(0.00)	11.43(0.00)	10.27(0.00)	8.66(0.00)	22.63(0.00)	21.82(0.00)	20.92(0.00)
	SCBGL('23)	13.27(0.00)	13.39(0.00)	14.62(0.00)	11.65(0.00)	11.36(0.00)	11.78(0.00)	23.05(0.00)	22.66(0.00)	23.21(0.00)
	DAQINT('24)	13.73(0.00)	11.25(0.00)	10.25(0.00)	11.54(0.00)	8.64(0.00)	7.66(0.00)	23.46(0.00)	20.21(0.00)	19.87(0.00)
	DCGA('25)	14.94(0.00)	14.22(0.00)	12.69(0.00)	15.63(0.00)	13.87(0.00)	11.48(0.00)	26.48(0.02)	25.09(0.00)	22.99(0.00)
	OIMC-IAGD('26)	16.75(0.89)	13.72(0.58)	12.48(0.68)	12.44(0.20)	9.99(0.20)	8.47(0.23)	23.09(0.62)	21.10(0.40)	20.28(0.40)
	PBDAL('25)	12.65(0.00)	11.00(0.00)	9.63(0.00)	10.47(0.00)	8.13(0.00)	7.24(0.00)	22.34(0.00)	20.62(0.00)	19.92(0.00)
	ASCR('25)	23.30(0.31)	29.74(0.36)	39.59(0.49)	27.67(0.13)	36.60(0.14)	49.13(0.16)	38.33(0.25)	46.57(0.26)	58.35(0.31)
	Ours	61.84(0.00)	52.01(0.00)	50.29(0.00)	67.45(0.00)	66.70(0.00)	63.48(0.00)	79.15(0.00)	72.73(0.00)	66.82(0.00)

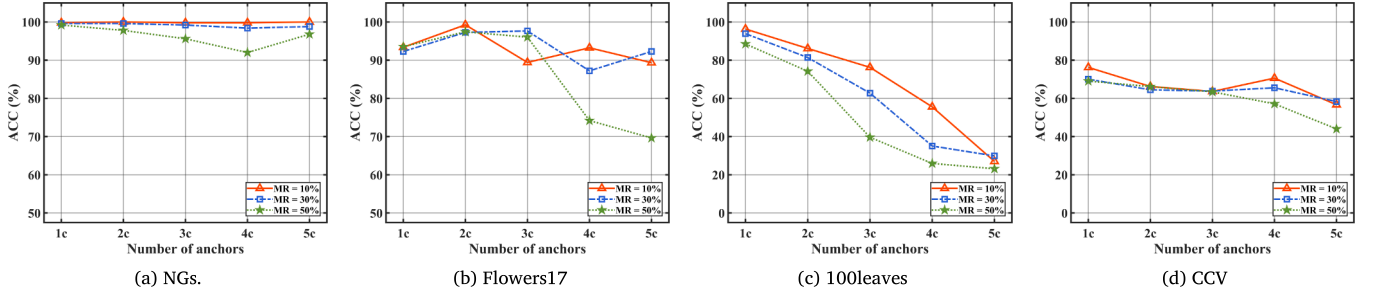


Fig. 2. The accuracy (ACC) performance of SIMC-ARTB across four datasets as the number of anchors varies under MR of 10%, 30%, and 50%.

- 2) **CCA-IMVC** (TCSVT'23) (Yu et al., 2023): It optimizes the aligned anchors in their original feature space via a shared graph structure to resolve anchor mismatching, achieving enhanced clustering performance.
- 3) **SCBGL** (TCSVT'23) (Zhao et al., 2023): It constructs a global-guided consistent anchor matrix and leverages both intra- and inter-view bipartite graphs to self-complete missing information.
- 4) **DAQINT** (TCYB'24) (Yu et al., 2024b): It predefines multiple anchor sets each view and employs adaptive weighting to approximate the optimal number, producing diverse-scale bipartite graphs.
- 5) **DCGA** (TNNLS'25) (He et al., 2025): It jointly models intra- and inter-view correlations through information bottleneck guided anchor learning and alignment, enabling anchors to retain essential knowledge.
- 6) **OIMC-IAGD** (KBS'26) (Wu et al., 2026): It unifies structure imputation and clustering within a framework by jointly performing imputed anchor graph learning and discrete label acquisition.
- 7) **PBDAL** (TMM'25) (Li et al., 2025): It adopts a paired and balanced dynamic anchor learning mechanism that supervises anchor consistency and fairness, coupled with an enhanced bipartite graph tensor learning strategy.
- 8) **ASCR** (AAAI'25) (Xu et al., 2025): It employs independent anchor learning and sample embeddings in a latent subspace to achieve reliable graph completion while preserving discriminative semantic information across incomplete views.

In all baseline methods evaluated in this study, we utilize the official implementations released by the original authors. To ensure a fair and reproducible comparison, we employ exactly the hyperparameter ranges reported as optimal in the original paper. Specifically, we vary λ in $[1e-4, 1e-2, 1e0, 1e2, 1e4]$, search for μ in $[1e-4, 1e-2, 0, 1e0, 1e2, 1e4]$, and tune the number of anchors in $[c, 2c, 5c]$ for SIMVC-SA. In CCA-IMVC, the number of anchors and λ are searched in $[c, 2c, 3c, 4c, 5c]$ and $[1e-2, 1e-1, 1e0, 1e1, 1e2]$, respectively. For SCBGL, we search λ_1 in the range $[1e-2, 1e-1]$, while λ_2 and λ_3 are chosen from $[1e-3, 1e-2, 1e-1, 1e0, 1e1, 1e2]$. In DAQINT, we tune the parameters λ in the range $[1e-2, 1e-1, 1e0, 1e1, 1e2]$ and β in $[1e2, 1e3, 1e4, 1e5, 1e6]$, respectively. In DCGA, we tune α and β within $[1e-3, 1e-2, 1e-1, 1e0, 1e1]$, and the number of anchors within $[c, 2c, 3c, 4c, 5c]$. In OIMC-IAGD, λ, β, γ are searched within $[1e-2, 1e-1, 1e0, 1e1, 1e2]$, while the number of anchors is selected from $[c+5, c+10, c+20, c+50]$. For PBDAL, we set β within $[2^{-5}, 2^{-3}, 2^{-1}, 2, 2^3, 2^5]$, γ in $[2^{-5}, 2^{-3}, 2^{-1}, 2, 2^3, 2^5]$, and m in $[2c, 3c, 5c, 7c, 10c]$. For ASCR, we search α in the range $[2^2 \sim 2^7]$.

For our SIMC-ARTB model, we note that the parameters α, β, γ and m correspond to the weighted alignment fusion term, the label graph learning term, the WTNN tensor regularization term, and the number of anchors, respectively. Specifically, we tune α within the range $[1e-2, 1e-1, 5e-1, 1e0, 1e1, 1e2]$, β within $[5e-5, 1e-4, 5e-4, 1e-3, 1e-2, 1e-1]$, and γ within $[5, 10, 50, 75, 100, 500]$. To simplify the anchor fine-tuning procedure, we further set the anchor number to $m = c$.

6.1.3. Evaluation metrics

We utilize three standard and widely accepted evaluation metrics to quantify the performance of the proposed SIMC-ARTB: Accuracy (ACC), Normalized Mutual Information (NMI), and Purity (PUR). Although each metric captures distinct characteristics of clustering quality, all are designed such that larger values correspond to better clustering performance. Furthermore, each method is executed independently ten times on every benchmark dataset.

6.2. Comparison results

The clustering performance of the proposed method and eight baseline approaches on eight incomplete multi-view datasets is reported in Table 3. For each method, we present the mean performance and standard deviation (denoted as $means(std)$) computed over ten runs, thereby ensuring statistical robustness and reproducibility of the findings. The best average performance on each dataset is highlighted in boldface, while the second-best results are underlined. From these comparison results, the following important observations can be made:

- Our proposed SIMC-ARTB consistently demonstrates superior clustering capability under varying missing ratios MR . Although it does not always rank first, it maintains a highly competitive performance. On Flowers17, 100leaves, CCV and NUSWIDE OBJ, it outperforms all competing methods in terms of ACC, NMI, and PUR. On the remaining datasets, it attains comparable performance, further highlighting the robustness and effectiveness of SIMC-ARTB.
- Among bipartite graph methods for IMC, tensor-based bipartite graph models like SIMC-ARTB, ASCR and PBDAL consistently outperform traditional bipartite graph formulations. Across most datasets, SIMC-ARTB achieves the highest clustering accuracy, with ASCR ranking second, demonstrating its superior trade-off between clustering accuracy and computational efficiency.
- Furthermore, even advanced one-step clustering frameworks, including the state-of-the-art OIMC-IAGD, fail to outperform traditional two-stage methods. This observation highlights the advantages of the proposed label graph learning scheme and demonstrates its strong capability for cluster discovery.
- For large-scale datasets such as SUNRGBD, Animal and NUSWIDE OBJ, which contain more than 10,000 samples, most competing methods exhibit noticeably degraded clustering performance. In contrast, our approach achieves highly promising clustering results, indicating that the proposed tensor-based bipartite graph framework is well-suited for large-scale IMC tasks. On the NUSWIDE OBJ dataset with 50% MR , our method achieves improvements of 10.7%, 14.35%, and 8.47% over the second-best method, ASCR, in terms of the three evaluation metrics.
- Effectiveness of anchor refactoring learning. The pseudo-label guided dynamic anchor learning method (PBDAL) achieves competitive clustering performance compared with many state-of-the-art approaches, particularly on small-scale datasets such as NGs and

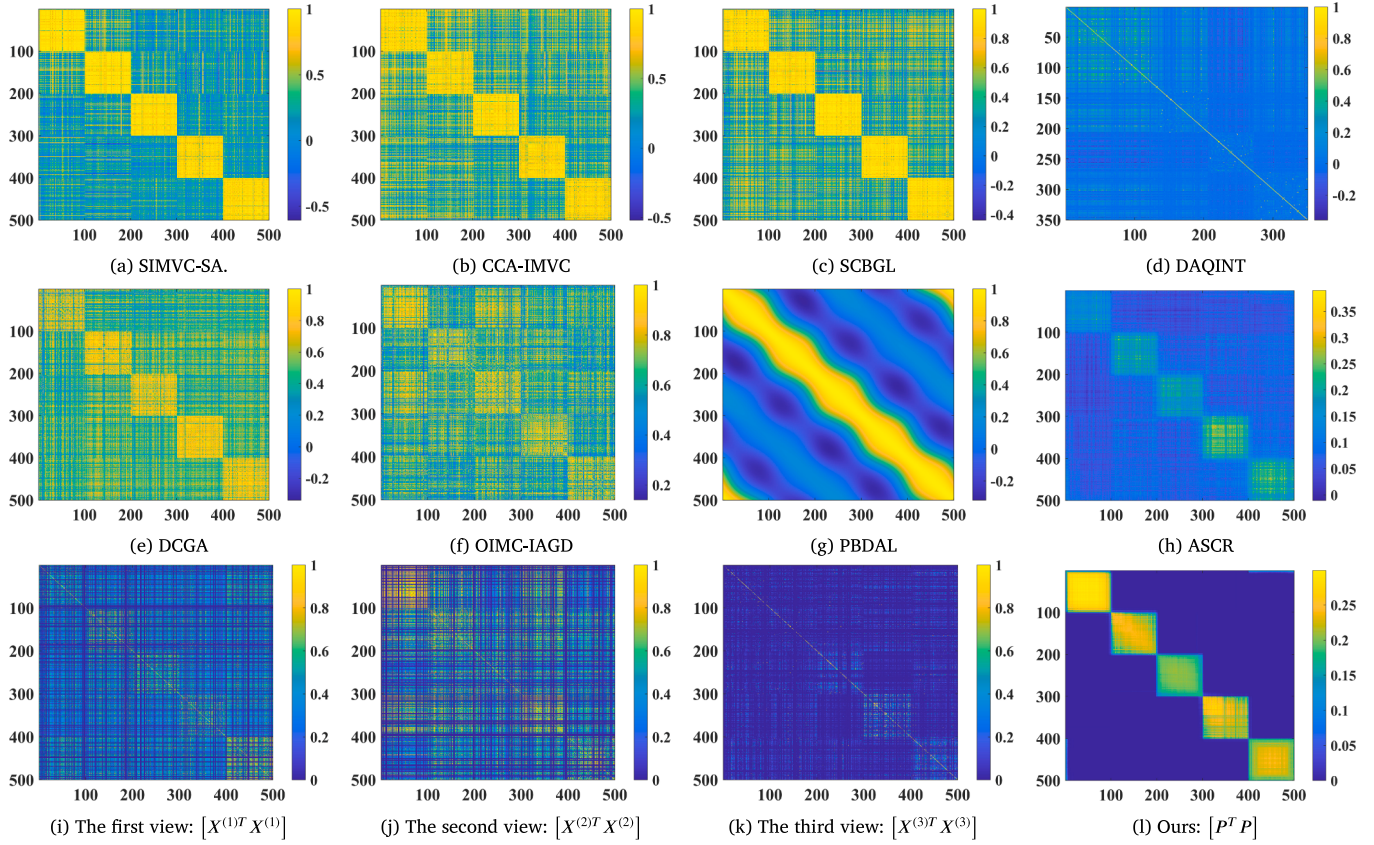


Fig. 3. The visualization of obtained correlations graph on the NGs with $MR = 30\%$.

100leaves. However, in most cases, its performance remains inferior to that of our SIMC-ARTB, which adopts an anchor refactoring learning strategy. This advantage arises because anchor refactoring allows the learned anchors to align more accurately with the underlying data distribution.

- It is worth noting that the clustering performance is not always monotonically correlated with MR . For example, similar performance can be observed on NGs, Flowers17 and SUNRGBD under MR of 10%, 30%, and 50%, whereas the performance on CCV and NUSWIDE OBJ varies considerably across different MR . Although a higher MR generally leads to greater information loss, the actual impact also depends on the structural importance of the missing samples. In addition, the proposed bipartite graph fusion module explicitly aggregates complementary structural information from multiple incomplete views, while the weighted bipartite graph tensor further exploits high-order, low-rank correlations to implicitly recover missing anchor relationships.

Collectively, SIMC-ARTB demonstrates superior performance across all evaluated datasets and maintains robustness when applied to incomplete multi-view data. These results clearly validate the effectiveness of the proposed SIMC-ARTB framework.

6.3. Ablation studies

In this section, to investigate the individual contributions of each component in the proposed model, we conduct a systematic ablation study. Four key design choices are analyzed, leading to the full SIMC-ARTB and its four ablated variants.

- 1) Removing the anchor refactoring (AR) term in Eq. (13) from SIMC-ARTB, denoted as w/o AR.

- 2) Removing the weighted alignment fusion (WAF) term in Eq. (23) from SIMC-ARTB, denoted as w/o WAF.
- 3) Removing the label graph learning (LGL) term in Eq. (28) from SIMC-ARTB, denoted as w/o LGL. In this case, cluster labels are generated using *Kmeans*.
- 4) Replacing the WTNN term in SIMC-ARTB with the standard nuclear norm, denoted as w/o WTNN.

The evaluation was performed on the NGs, 100leaves, CCV, and SUNRGBD datasets under various MR to simulate real-world incomplete data scenarios. Performance was measured using the average clustering accuracy and the corresponding standard deviations across ten experimental runs. As shown in Table 4, several observations can be drawn:

- The full version of the SIMC-ARTB model, which incorporates all four designed components, exhibits the highest and most stable ACC results across different datasets and missing ratios. These results indicate that the superior performance arises from the synergistic effect of all components rather than the dominance of a single one. In particular, on the CCV dataset, the model achieves an ACC of approximately 67.03% even under a high missing rate ($MR = 50\%$), demonstrating strong robustness to view incompleteness.
- The exclusion of any individual component leads to a noticeable drop in clustering accuracy. Notably, the w/o WTNN variant shows consistently inferior performance across all datasets, highlighting the importance of the WTNN component in handling IMC tasks, where it plays a decisive role in enhancing clustering quality. While the w/o AR model delivers relatively favorable outcomes on some datasets, the complete SIMC-ARTB outperforms it by significant margins, including improvements of 7.38% on 100leaves ($MR = 30\%$), 8.71% on CCV ($MR = 30\%$) and 4.34% on SUNRGBD ($MR = 30\%$).
- The w/o LGL variant employing *Kmeans* clustering produces only moderate and unstable performance, as reflected by the increased

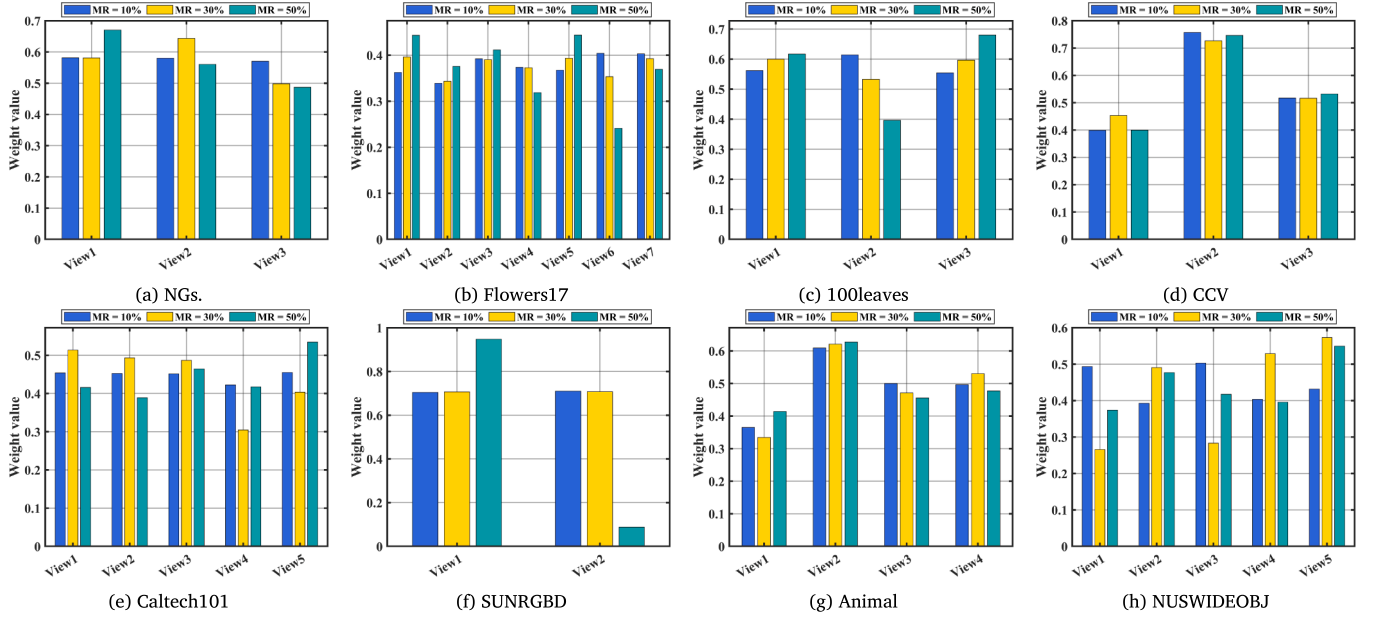


Fig. 4. Visualization of the learned adaptive weighting vector ω across eight datasets.

Table 4

Ablation study evaluating core components of the proposed method with ACC(%) [means(std)].

MR	Ablation methods	NGs	100leaves	CCV	SUNRGBD
10%	w/o AR	97.40(0.00)	90.38(0.00)	69.08(0.00)	22.45(0.00)
	w/o WAF	98.80(0.00)	86.44(0.00)	42.98(0.00)	26.78(0.00)
	w/o LGL	97.40(0.00)	69.03(2.43)	64.22(2.43)	20.93(0.63)
	w/o WTNN	92.20(0.00)	53.56(0.00)	20.79(0.00)	21.37(0.00)
	Ours	99.80(0.00)	96.31(0.00)	76.23(0.00)	27.05(0.00)
30%	w/o AR	98.80(0.00)	86.50(0.00)	61.35(0.00)	22.05(0.00)
	w/o WAF	85.40(0.00)	88.81(0.00)	36.32(0.00)	25.32(0.00)
	w/o LGL	87.02(0.20)	77.73(1.17)	56.90(0.76)	21.87(0.67)
	w/o WTNN	82.40(0.00)	39.13(0.00)	19.34(0.00)	19.58(0.00)
	Ours	99.60(0.00)	93.88(0.00)	70.06(0.00)	26.39(0.00)
50%	w/o AR	98.40(0.00)	86.44(0.00)	63.12(0.00)	23.27(0.00)
	w/o WAF	86.00(0.00)	78.19(0.00)	22.93(0.00)	20.52(0.00)
	w/o LGL	69.62(0.06)	74.70(1.48)	35.17(2.94)	19.24(0.74)
	w/o WTNN	68.80(0.00)	28.13(0.00)	18.09(0.00)	11.92(0.00)
	Ours	99.20(0.00)	88.50(0.00)	67.03(0.00)	23.62(0.00)

variance. This demonstrates that incorporating label graph learning into the tensor-based bipartite graph model not only enhances clustering performance but also improves model stability.

- The ablation results clearly demonstrate that each module in the proposed framework plays a critical role in the overall model performance. Their joint integration proves indispensable for achieving high-accuracy IMC clustering results.

6.4. Anchors number studies

We investigate the influence of the number of anchors on SIMC-ARTB by varying m within $[1c, 2c, 3c, 4c, 5c]$, where c denotes the number of clusters. The clustering ACC is evaluated on four datasets, and the results are shown in Fig. 2. For NGs, 100leaves, and CCV, the best ACC is consistently obtained when $m = c$ under multiple missing rates. For Flowers17, the optimal result occurs at $2c$, while the ACC still exceeds 90% when $m = c$. Although increasing the number of anchors generally improves the representation capacity of conventional anchor-based methods, such improvements mainly originate from enhanced feature space coverage rather than improved anchor quality. In the proposed SIMC-

ARTB, anchor refactoring is introduced to learn representative anchors that accurately preserve the underlying data structure. Consequently, a relatively small number of high-quality anchors is sufficient to characterize the data distribution. When the number of anchors becomes excessively large, redundant anchors may be introduced, increasing noise and weakening the discriminative ability of the anchor graph. As a result, the clustering performance may gradually deteriorate. Moreover, increasing the number of anchors enlarges the bipartite graph tensor and requires additional anchor fine-tuning, leading to higher computational cost and reduced scalability. Thus, setting $m = c$ usually yields satisfactory clustering performance while maintaining computational efficiency.

6.5. Visualization of consistent affinity graph

To verify the performance of SIMC-ARTB in recovering graph information and discovering cluster structures, we visualize the affinity graphs to illustrate their block-diagonal characteristics. These visualizations are obtained from the NGs dataset with a missing ratio of 30%, where $X^{(v)T}X^{(v)}$ is computed from incomplete feature sets and $P^T P$ is reconstructed by SIMC-ARTB together with eight baseline models for comparison. As depicted in Fig. 3, the three-view affinity graphs under real-world missing conditions exhibit significant structural degradation, which poses challenges for conventional anchor graph fusion-based imputation methods in capturing inter-view consistency and complementarity. Then, a comparison between the missing affinity graphs in Fig. 3 (i)–(k) and the graph recovered by the tensor-based SIMC-ARTB in Fig. 3 (l) reveals that the complete block-diagonal structure inherent in the original data is successfully reconstructed. This result verifies the effectiveness of the proposed method in recovering incomplete graph information. Finally, we compared the affinity graphs obtained by eight baseline algorithms and SIMC-ARTB. As shown in Fig. 3 (a)–(h), the affinity graphs generated by SIMC-ARTB exhibit clearer block-diagonal structures and more consistent connectivity patterns compared with those produced by the baselines. These findings demonstrate that our method is effective not only in recovering structural information but also in capturing consistent correlations among incomplete multi-view.

6.6. Visualization of weights

We visualize the learned adaptive weighting vector ω on eight datasets with the number of views ranging from 2 to 7, as shown

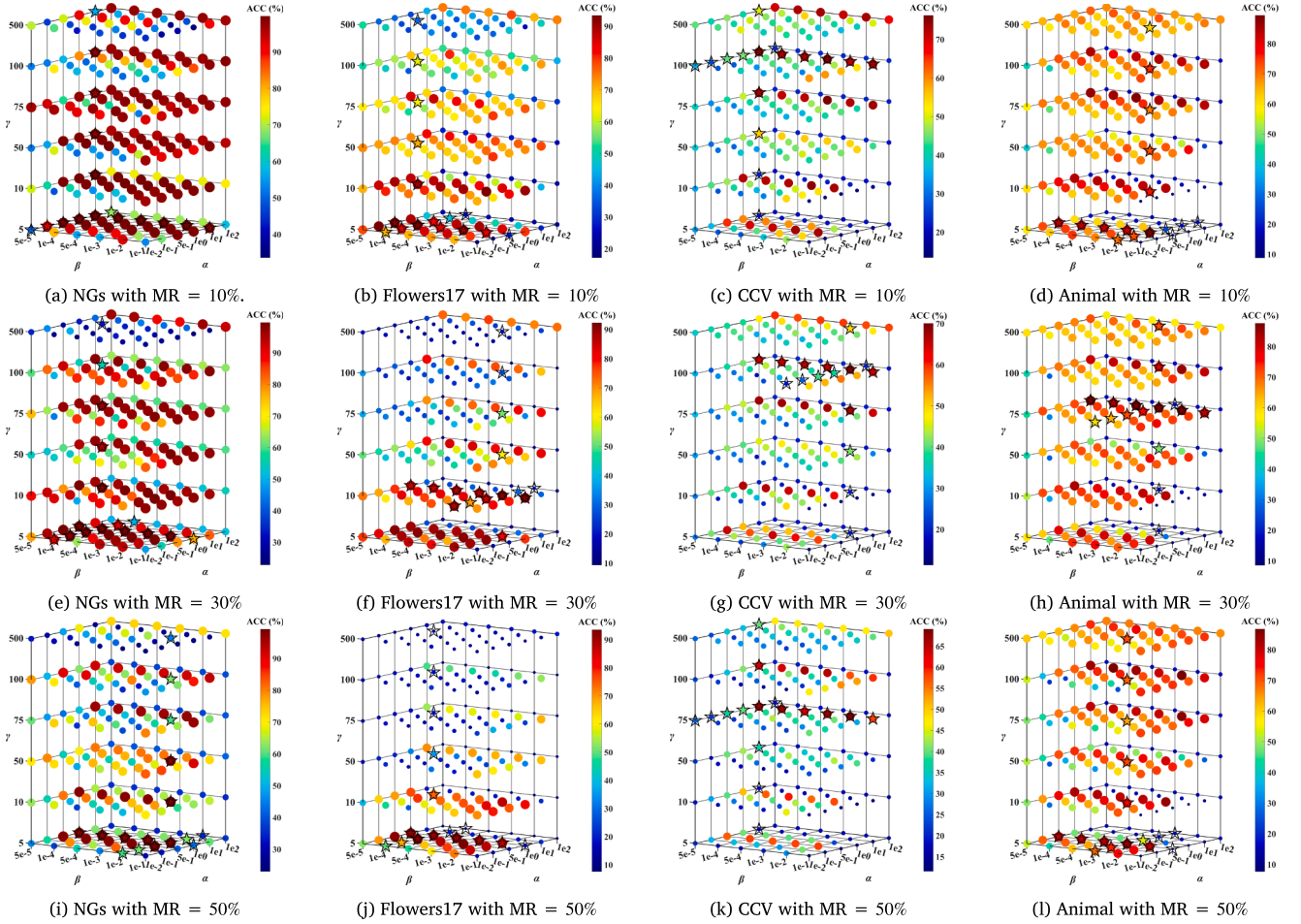


Fig. 5. Parameter sensitivity analysis on NGs, Flowers17, CCV and Animal datasets in terms of ACC (%). ★ represents the best hyperparameter search in experiments.

in Fig. 4. It can be observed that the learned weights exhibit significant variations across datasets with different numbers of views, such as Caltech101 with 5 views and SUNRGBD with 2 views. In addition, for the same dataset under different MR , the weights assigned to the same view also change accordingly, as observed on the Flowers17 and NUSWIDE OBJ datasets. These findings reveal that the adaptive weighting strategy effectively accounts for both inter-view importance differences and the influence of missing samples during the fusion of multiple incomplete bipartite graphs.

6.7. Hyperparameters sensitivity investigation

In this section, we conduct a sensitivity analysis of the clustering performance with respect to the hyperparameters α , β , and γ . A series of comparative experiments are performed under multiple MR on NGs, Flowers17, CCV and Animal datasets. In order to provide an intuitive visualization of the joint influence of these three hyperparameters, we map them into a three-dimensional parameter space, with model accuracy ACC (%) encoded jointly by the color and size of the data points (see Fig. 5). The star-marked bubble plots reveal insights into parameter sensitivity. For NGs dataset Fig. 5 (a), (e), (i), the model exhibits notable sensitivity to changes in α , with optimal performance achieved when α lies between 1 and 10 under all MR settings. By contrast, β and γ show broader tolerance ranges, with β performing best within $5e-5$ to $1e-2$ and γ within 5 to 100. For the CCV dataset Fig. 5 (d), (h), (l), a markedly different pattern emerges. The hyperparameter β exerts only a minor influence, with near-optimal clustering accuracy preserved across the broad interval $5e-5$ to $1e-2$ for all missing rates. Although higher

β values (e.g., $\beta = 1e-1$) slightly affect the results, their influence remains limited. In contrast, both α and γ play more influential roles: peak performance is observed when α is around 10, while the most favorable outcomes for γ occur within the range from 75 to 500.

6.8. Convergence analysis

As shown in Fig. 6, the convergence behavior of SIMC-ARTB is examined on NGs, Flowers17, CCV, and Animal datasets under missing rates of [10%, 30%, 50%]. It can be observed that the ACC values increase monotonically and eventually stabilize at high-performance levels, while the Reconstruction Errors (RE) decrease sharply and gradually become stable within a limited number of iterations. Both curves reach stable plateaus thereafter, clearly indicating that SIMC-ARTB achieves convergence efficiently.

6.9. Execution time comparison

To assess the efficiency of SIMC-ARTB, we illustrate how the running times of multiple IMC algorithms vary as the dataset size increases from 500 to 30,000 in Fig. 7. As observed, tensor-free models follow a linear time complexity $\mathcal{O}(n)$, whereas SIMC-ARTB demonstrates the same $\mathcal{O}(n \log n)$ complexity as the graph tensor-based ASCR and PBDAL. In comparison with several existing approaches, including bipartite graph methods CCA-IMVC, DAQINT, DCGA, tensor-based IMC (PBDAL), and one-step clustering algorithms OIMC-IAGD, SIMC-ARTB achieves improved computational efficiency maintaining strong clustering performance (e.g., on CCV, Animal and NUSWIDE OBJ datasets). While the

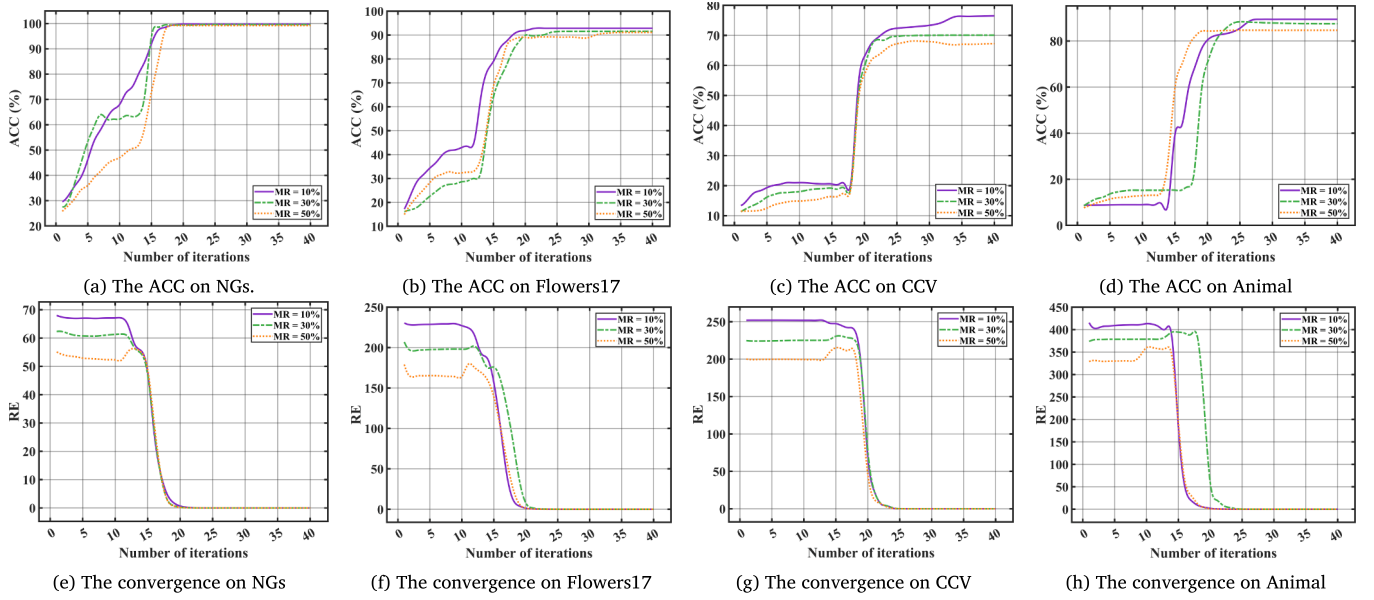


Fig. 6. The convergence performance analysis of NGs, Flowers17, CCV and Animal.

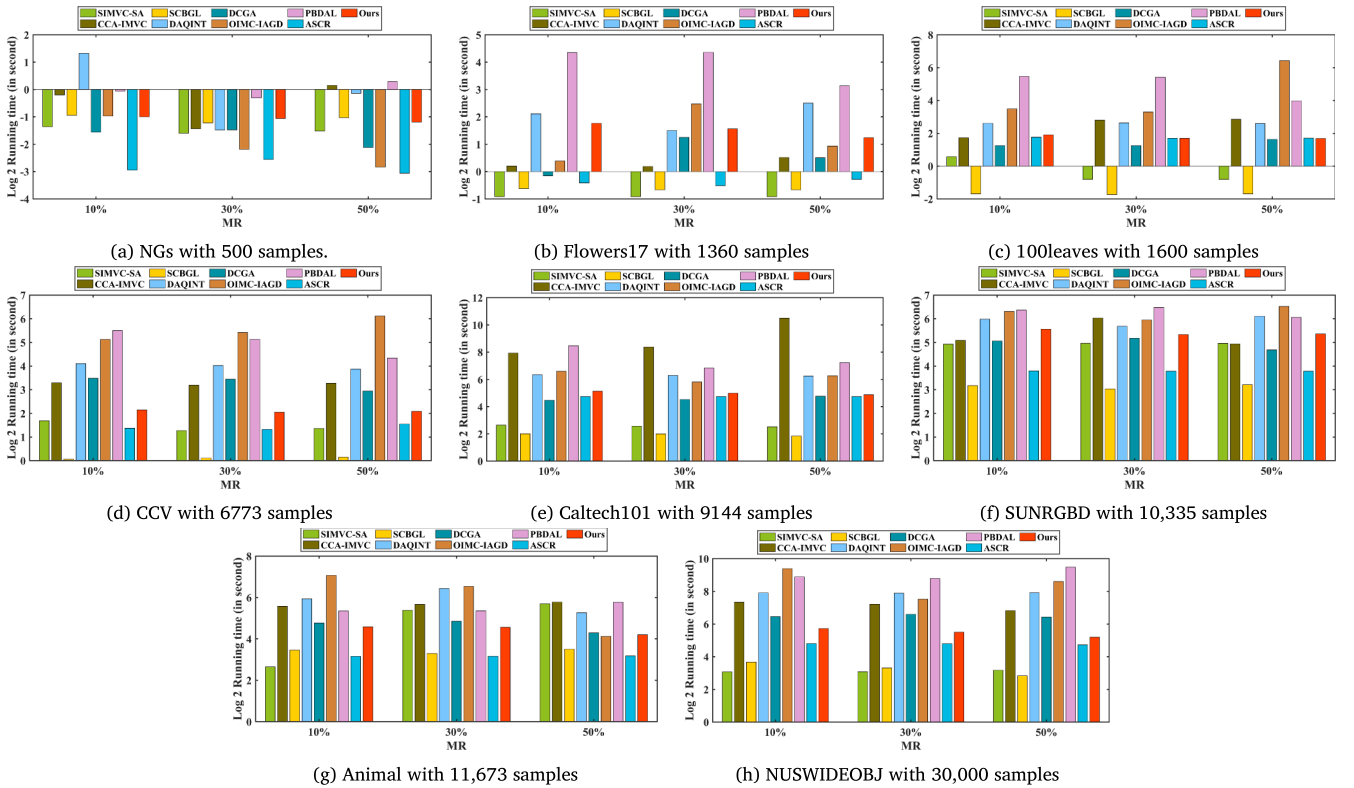


Fig. 7. Relative running time comparison (logarithmic scale) among nine algorithms over eight datasets.

SIMVC-SA, SCBGL and ASCR demonstrate faster execution speeds, their performance is limited by the quality of the constructed bipartite graph. This is because SIMC-ARTB integrates the WTNN constraint on the bipartite graph tensor, which requires SVD operations and the optimization of tensor Lagrangian multipliers, thereby increasing computational cost. Nevertheless, SIMC-ARTB achieves an effective balance between clustering accuracy and computational efficiency, demonstrating its suitability for large-scale IMC tasks.

7. Conclusion

In this paper, we propose SIMC-ARTB, a novel and comprehensive framework for addressing incomplete multi-view clustering (IMC) from a bipartite graph clustering perspective. Specifically, our framework constructs an incomplete bipartite graph for each view by leveraging refined anchors through an anchor refactoring strategy. Furthermore, a weighted alignment fusion module and a tensorial bipartite graph

component are employed to jointly recover a reliable and consistent bipartite graph across multiple views. Meanwhile, label graph learning is seamlessly integrated into the unified framework, providing a direct pathway for cluster label generation. Extensive experiments conducted on eight benchmark incomplete datasets under various missing ratios demonstrate both the effectiveness and efficiency of the proposed method. In future work, we will investigate adaptive anchor configuration strategies and develop a more principled guideline for anchor selection. The proposed method still relies on a predefined number of anchors, which is selected empirically in the current study. While representative anchor learning alleviates the dependence on large anchor sets, the relationship between anchor quantity and the intrinsic cluster structure has not been explicitly explored.

CRedit authorship contribution statement

Xuan Chen: Writing – original draft, Methodology, Data curation, Conceptualization; **Zhikui Chen:** Writing – review & editing, Supervision, Funding acquisition; **Enze Ji:** Writing – review & editing, Visualization; **Chuan Xiao:** Supervision, Funding acquisition; **Liang Zhao:** Writing – review & editing, Supervision; **Xiaodi Huang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

Data are available upon reasonable request.

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